

Synergies and Gaps in Support for Technical Career Development in UK Higher Education and Research

A Semi-Quantitative Analysis of 'Technician Commitment' Action Plans and Progress Reports using Natural Language Processing

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Abstract

This study analyses Technician Commitment action plans from UK higher education institutions to address persistent technical talent demand in research. Using natural language processing with Google Gemini, 3558 action items from 104 plans were categorised into 26 key themes. The most common focus areas were "Ongoing Visibility and Communication", "Networking and Presenting" and "Career Pathways and Progression", whilst "Wellbeing and Mental Health", "Environmental Sustainability" and "Equipment Sharing" received minimal attention.

Comparison with the TALENT Policy Commission's recommendations revealed strong alignment on improving visibility and diversifying entry routes, but significant gaps in technical staff's policy involvement and strategic workforce planning. Notably, institutions rarely cited external resources from professional bodies, suggesting underutilisation of available support.

The study proposes developing a 'Skills Portfolio' methodology with a standardised 'living skills ontology' to provide verifiable mechanisms for demonstrating technical expertise. These findings highlight the need for more prescriptive national guidance and targeted outreach to foster comprehensive technical workforce development.

Keywords

Technician Commitment

Technical workforce development

Higher education policy

Career pathways

Research infrastructure

Machine learning classification

Natural language processing

Introduction

Multiple governments and ministers from across the political spectrum have positioned the UK as a world leader, both a) in science, research and innovation (CaSE, 2023; Professional Engineering, 2023; UK Department for Science, Innovation and Technology, 2025; UK Government, 2023; Weston, 2023) driven through the activities of higher education and research institutions, start-ups and small/medium business enterprises (CaSE, 2024; Castell, 2025; West, 2024); and b) as a biomedical enabler through harnessing the NHS and digital transformation to deliver clinical research (Jacobs, 2025; NHS England, 2025; UK Government, 2021, 2025a). The UK's Industrial Strategy Invest 2035 (UK Department for Business and Trade, 2024) and Modern Industrial Strategy policy paper (UK Government, 2025b) outline areas where the government plans for growth in the coming 10 years, including advanced manufacturing, clean energy industries, creative industries, defence, digital and technologies, financial services, life sciences and professional and business services. Staff working in diverse technical capacities in a range of different sectors underpin UK research and development and are key to the delivery of research outputs and impactful products (BBSRC, 2023; The Royal Society, 2021; Vere, 2021). Training and development in vocational and academic learning environments feeds directly into the industrial scientific output of 'UK PLC' and enables both public services and new impactful businesses to succeed (AdvanceHE, 2024; Greaves and Brawley, 2025; HESA, 2024; Universities UK, 2024).

To address the need to develop technical talent to drive scientific R&D in the UK, extensive engagement has been used to build and develop technical staff communities, and define and disseminate the Technician Commitment (ITSS, 2025b). Subsequent work culminated in publication of the TALENT Policy Commission Report (Midlands Innovation, 2022). In 2023 the UK Institute for Technical Skills and Strategy (ITSS) was formed (Thorne, 2023) (ITSS, 2025a), which is continuing to implement initiatives to support technical staff in UK higher education and research institutions including acting as the new home of the Technician Commitment (Technician Commitment, 2024). The Commitment was developed to improve "...visibility, recognition, career development and sustainability for technicians working in higher education and research, across all disciplines" (ITSS, 2025b). Signatories are invited to define and publish action plans containing specific initiatives and actions which they will use to address these four pillars of the Commitment. These are aligned with the TALENT Policy Commission recommendations (Midlands Innovation, 2022) which outline 16 overarching recommendations plus additional recommendations specific to different stakeholder sectors.

Multiple reports argue that technicians in UK Universities and Research Institutes often take on additional responsibilities in terms of service delivery, teaching or research which are not formally recognised (Noke et al., 2024; Vere and Harris, 2022; Wragg et al., 2023).

Recognition of the full contribution of the role of technical staff in research, teaching or knowledge exchange environments can be lacking or hard to achieve, sometimes due to

institutional barriers (Vere et al., 2024); this has potentially negative impacts at several levels, and may affect the career development of an individual or the accurate appreciation of skills within a workforce, undermine the professional appeal of technical careers and cause the loss of business-critical technical skills as technicians switch to roles where their skills can be better recognised.

Because “technical” can encompass a diverse range of roles, it has been difficult to establish a clear, consistent definition of this staff group (Ivie, 2019). An early definition came from the Science Council (Science Council, 2017), and recently, HESA and ITSS proposed the following definition for Technical Staff in the context of UK higher education and research institutions (Van Essen-Fishman, 2025): “Technical staff are those who are employed to use specialised knowledge of techniques, tools, and technology. They make practical contributions to teaching and learning, research, innovation, and enterprise activities... ”.

The aim of this study was to analyse the actions pledged by higher education and research institutions in the UK as a measure of workforce and strategic planning in the area. To do so, all Technician Commitment Action Plans published by UK higher education and research institutions were examined at an individual ‘Action’ level using the Python- and Google Gemini-enabled workflow outlined in Figure 1. All Actions were extracted from all available Action Plans, and natural language processing and human-in-the-loop clustering were used to classify actions in themes and enumerate them. Subsequently, a similar workflow was built to classify and analyse the results of retrospective Red-Amber-Green (RAG) analyses of pledged tasks. Resources available from other sources in the UK landscape intended to meet those themes were identified and listed with categories, and compared with actions to identify coverage. Pledged actions were also aligned with the TALENT Policy Commission recommendations, to identify which recommendations would be addressed by the planned actions. Finally, gaps in the provision landscape and the language around technical skills are discussed, including how this analysis highlights opportunities to develop infrastructure to help connect and nurture the existing world-class technical capabilities in the UK.

Methods

a. Overview

Action plan documents were analysed from UK higher education and research institutions that have signed the Technician Commitment, examining what institutions committed to doing, how they categorised their actions, and what progress they reported. The full analytical workflow is shown in Figure 1, and detailed technical methods are provided in the Supplementary Methods). All code and data is available at https://github.com/Sammijj/TCDOC_EXTRACT.

This study conforms to the following reporting guidelines: CARDA - Guiding Document Analyses in Health Professions Education Research (Cleland et al., 2023); TRIPOD-LLM - Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis - Large Language Models (Gallifant et al., 2025); SRQR - Standards for Reporting Qualitative Research (O'Brien et al., 2014); and JARS-Qual - Journal Article Reporting Standards for Qualitative Research (Levitt et al., 2018).

b. Analytical Framework

This analysis treats Technician Commitment action plans as transparent records of institutional intentions while recognising they are socially constructed documents reflecting institutional priorities, resources, and policy contexts. Documents were analysed for stated commitments (what institutions pledge to do) with the understanding that they represent aspirational plans at a specific time point rather than completed actions. The analysis is situated within a broader policy ecosystem, comparing institutional plans against external resources and national policy recommendations to identify synergies and gaps in the technical workforce development landscape.

c. Document Collection

Action plan documents were collected from higher education and research institution websites and the Technician Commitment website (see Table S1 for full list). Where institutions had produced multiple versions, only the most recent action plan was used. Documents were stored and organised systematically to enable automated processing. Documents were collected between 1/8/25 and 31/8/25 and comprised publicly available policy documents authored by institutional staff (typically equality/diversity officers, technical services managers, or research culture leads) for accountability to the Technician Commitment initiative. Action plans varied substantially in format (structured tables versus

narrative text), length (8-77 actions per plan, mean 35 ± 17), and specificity of commitments, but all represented institutional intentions for technical staff development.

d. Extracting Actions from Documents

Automated text extraction was used to identify specific action items, tasks, or commitments from each document. A machine learning model (Google Gemini 2.5) was instructed to extract all discrete actions and present them as a structured list. This approach allowed consistent processing across documents of varying formats and lengths.

e. Categorising Actions

A system to classify actions into meaningful themes was developed. Starting with nine broad categories based on common themes in workforce development (such as recognition, training, and career progression; Table S2), a dictionary of keywords was built associated with each category. The keyword lists were refined through multiple rounds of testing until the system reliably identified action types (Table S3).

Each action was then scored against all categories based on which keywords it contained. Actions could be assigned to multiple categories if relevant, or flagged as uncategorised if they did not match any theme strongly enough. The categorisation system's accuracy was tested against a manually classified sample of 341 actions (~10% of the total), achieving good sensitivity (correctly identifying relevant actions) and specificity (correctly excluding irrelevant actions) across most categories (see Table 1).

f. Identifying Sub-themes Within Categories

To understand the diversity of approaches within each broad category, more specific sub-themes were identified. A statistical method (term frequency and inverse document frequency - TF-IDF) was used to find the most distinctive and important words within each category. Based on these results, Google Gemini defined sub-themes representing different aspects of each theme. Actions were then assigned to sub-themes using the same keyword-matching approach as the main categorisation. Categories with 100 or more occurrences in the dataset were used for further analysis in g below (n=8 categories).

g. Analysing Reported Progress

To track how institutional commitments evolved over time, an automated workflow was developed to analyse actions from both Action Plans and their corresponding Red/Amber/Green (RAG) assessment documents across multiple assessment periods. This analysis focused on understanding action progression between periods, where red indicated delayed or incomplete actions, amber indicated work in progress, and green indicated completed actions.

Action Extraction and Document Processing

All available Action Plan and RAG assessment documents were accessed via Google Drive. For documents with structured tables containing explicit RAG columns, intelligent column detection algorithms were used that could identify and merge continuation rows across multiple table sections. For unstructured documents, Google's Gemini 2.0 Flash language model was used to extract individual actions while preserving their semantic completeness. All extracted actions were assigned to the same 26-category taxonomy using the keyword-matching approach described above.

Action Matching and Trajectory Analysis

A three-tier matching strategy was implemented to track how actions evolved across assessment periods. First, for actions from the same period (e.g., comparing an Action Plan to its corresponding RAG assessment), TF-IDF similarity was again used to identify near-identical or closely matching actions. Second, for cross-period comparisons (e.g., tracking a RAG-assessed action from Period 1 to the next Action Plan in Period 2), Gemini AI batch processing was employed to perform intelligent semantic matching. The AI classified relationships as "Identical" (unchanged), "Extended" (more ambitious), "Narrowed" (more focused), "Related" (connected but different), or "Unrelated". This approach enabled us to distinguish genuine action evolution from simple continuity. Finally, trajectory codes were generated tracking whether each action Continued, was Related to a successor, Stopped with no continuation, or appeared as New with no predecessor.

Composite RAG-Trajectory Metrics

To understand how institutions responded to their commitments over time, four complementary metrics were developed that combine action completion status with evolution patterns. Each action in the RAG (Red-Amber-Green) review periods was assigned a composite code combining its completion status (Green = completed, Amber = ongoing, Red = delayed) with how it evolved into the next action plan (Continued, Related/adapted, or

Stopped). For example, a completed action that was sustained receives code "GC" (Green-Continued), while an ongoing action that was refined receives "AR" (Amber-Related).

From these composite codes, four metrics were calculated for each action category:

- **Successful, completed (%)**: The percentage of reviewed actions that were completed (Green status), measuring overall delivery on commitments.
- **Successful, continued (%)**: Among completed actions, the percentage that were sustained or evolved rather than stopped, measuring whether successful work is embedded into ongoing practice.
- **Adaptability score (0.0-1.0)**: Among challenged actions (Amber or Red status), the proportion that were refined or adapted rather than continued unchanged or abandoned, measuring institutional capacity to learn from difficulties.
- **Resilience score (0.0-1.0)**: Among challenged actions, the proportion that were sustained despite difficulties, measuring institutional commitment to maintaining difficult work.

These metrics were designed to be interpreted together. For instance, high completion with high continuity suggests strong institutional capacity to deliver and sustain work, while low completion with high adaptability indicates a learning-oriented approach that actively refines struggling initiatives. This framework enabled identification of distinct institutional patterns in how organisations approach and evolve their commitments.

h. Mapping External Resources

The support and resources available to technical staff beyond their own institutions were explored. Using the eight most common action sub-themes as examples, relevant UK-wide resources, professional bodies, networks, and support structures were systematically searched for. Higher Education and Research Institution's action plans were examined to determine whether they referenced these existing resources, providing insight into how well institutions were connecting their commitments to established infrastructure.

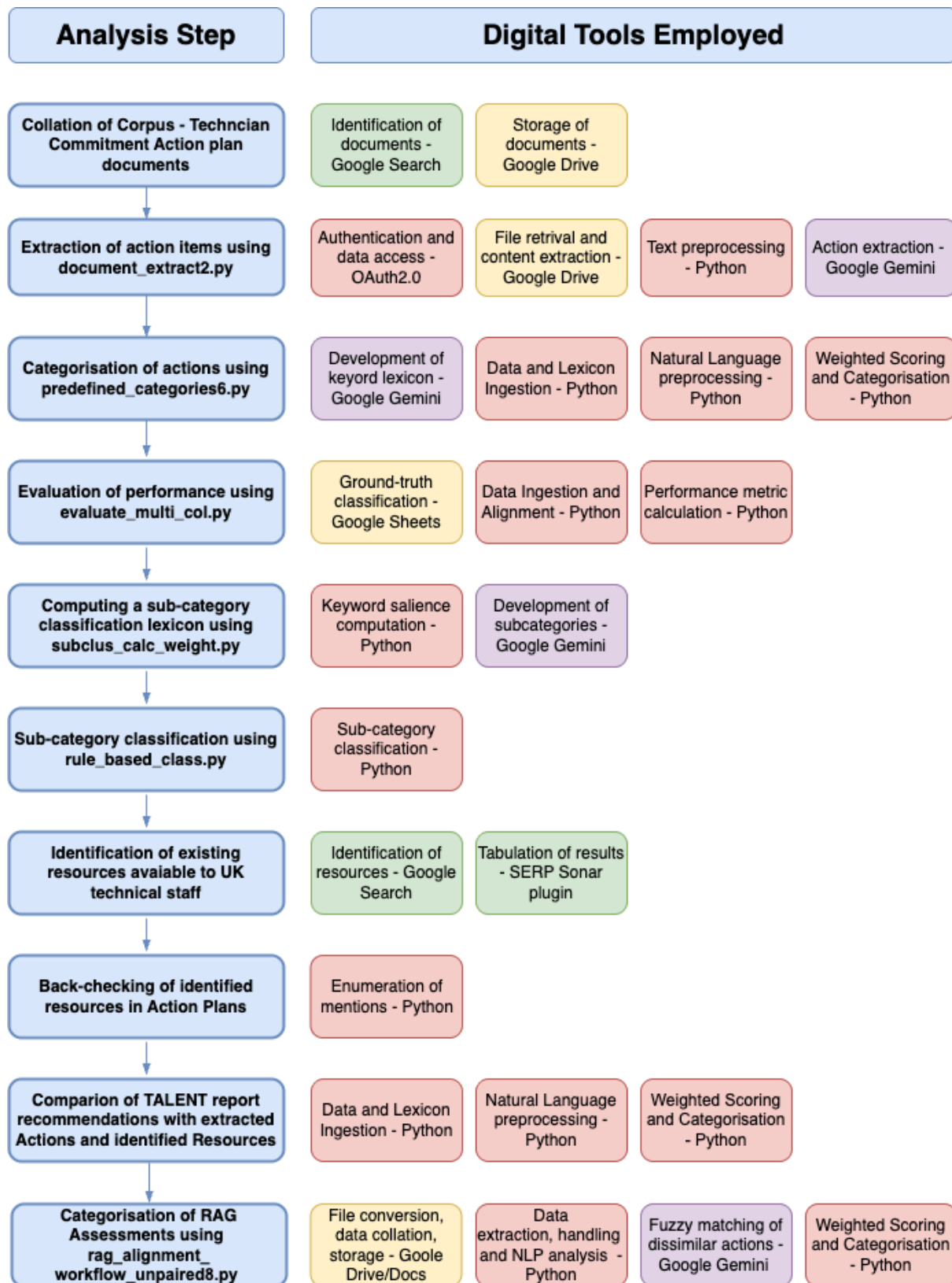
i. Alignment with National Recommendations

The actions in Higher Education and Research Institution action plans were compared against the 16 recommendations from the TALENT Policy Commission report, identifying

which national priorities were well-represented in institutional commitments and which received less attention.

Figure 1 - Analysis Workflow Overview: The analysis was carried out in a step-wise manner (blue) to enable iteration on specific steps and ensure reproducible results. Following collection of the corpus, a series of python scripts was developed to carry out each subsequent step, through extraction of the actions, categorisation, performance evaluation (sensitivity and specificity) and sub-categorisation. Subsequent timeframe and comparative analyses based on RAG assessments drew from these methods. Key to digital tools: Green - Google Search; Yellow - Google Drive; Red - Python code; Purple - Google Gemini.

Alt Text - The figure shows the workflow of the study as a series of differently coloured boxes, describing which of the types of tools and which code snippet was used for each stage.



Results

a. Corpus, extracted actions and ground truth - descriptive statistics

104 files PDF and Google Doc files were collated (Table S1) and analysed in this exercise. 3558 actions in total were extracted from the action plans by the Gemini-2.5-Pro model in around 55 minutes. This equated to an average of 35 (StDev = 17, range = 8-77) actions per action plan. After the categorisation, 141 actions (3.9%) remained uncategorised leaving a total of 3417 actions. A random sample of 10% (n=341) of categorised actions was used for blinded manual labelling to create the ground truth document for performance evaluation calculations.

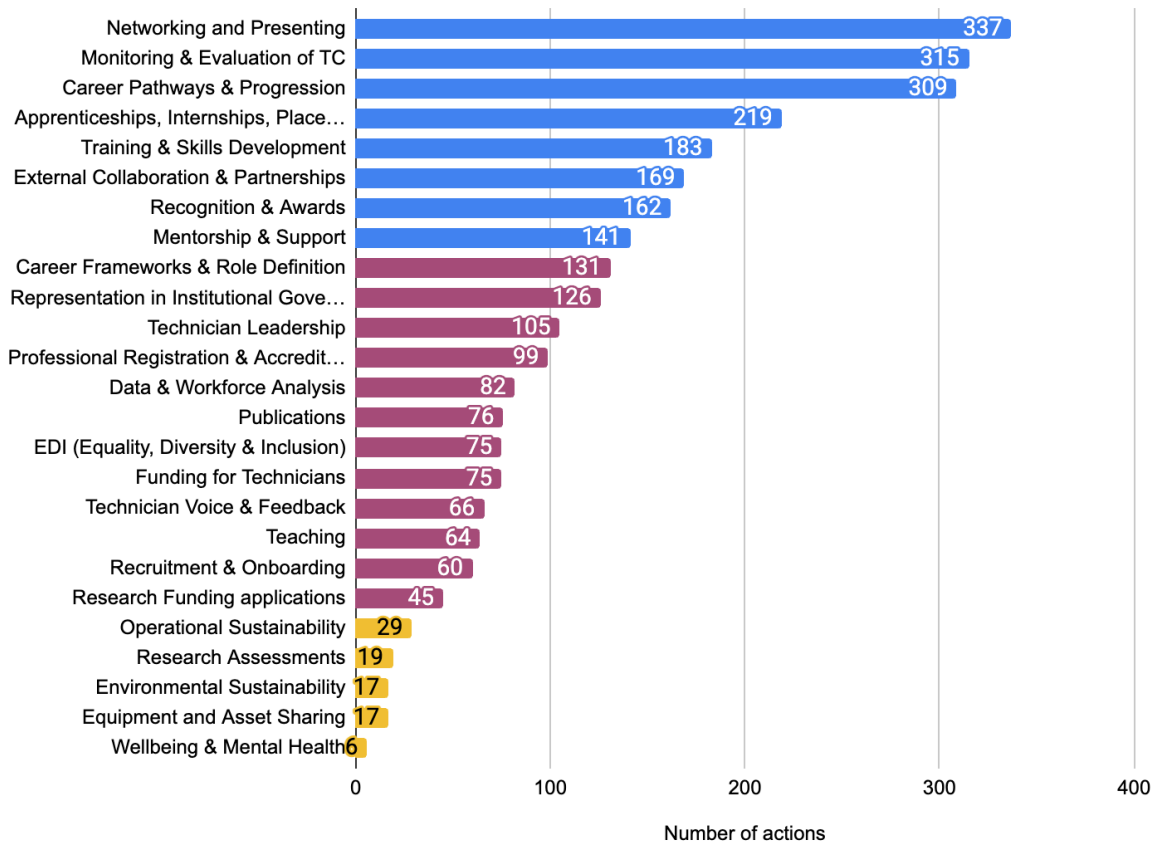
b. Categorisation of actions

The analysis defined 26 key categories of actions taken from the Actions Plans (see Table S3), the number of actions mentioned in a category shown in Figure 2a and the number of plans which mention an action at least once in Figure 2b. These were divided into arbitrary high (150+), medium (50-150) and low (below 50) brackets.

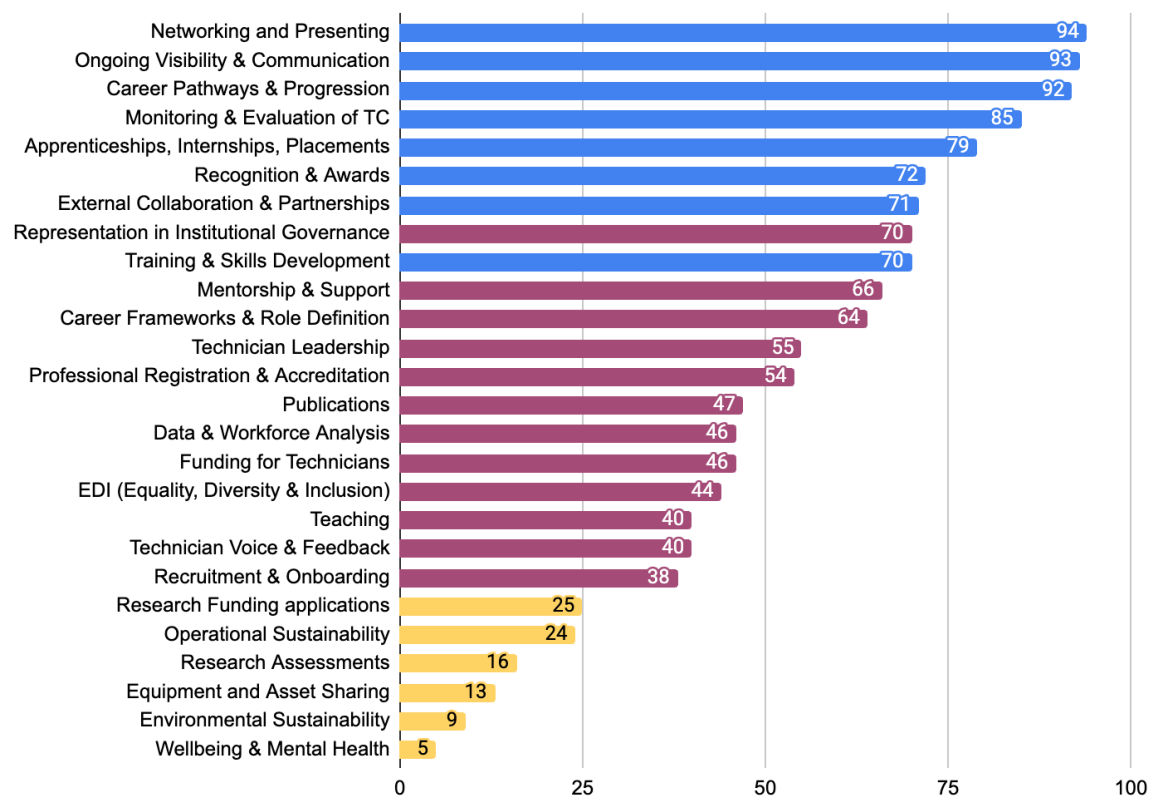
Figure 2 - Categorised Actions: Actions extracted from the action plans were categorised into one of 26 categories and enumerated. Categories with a higher number of actions indicate a high degree of engagement with this subject area. A: count of actions in each category; B: number of action plans which mention a category once. Colour bandings illustrate the low, medium or high engagement with the topic.

Alt Text - Two graphs show coloured bars enumerating the actions by count and mentions.

A. Number of actions classified in each category



B. Number of action plans which mention a category



c. Performance of categorisation

The performance of each category vs the ground truth is a key way to identify reliability of the extracted action numbers. As can be seen in Table 1, the macro-averages from the analysis were: Sensitivity (Recall) - 0.97; Specificity (Precision) - 0.78; F1 score - 0.85. Most categories displayed very high specificity (0.9 or above) indicating that most of the true negative values had been correctly identified. Sensitivity was more variable, with some categories falling below 0.6. However, many categories performed well, indicating that true positive values had been correctly attributed. Low-performing categories were found to often include Actions with more generalisable language or exhibit more potential overlap with other categories. Representation of the categories in the dataset also varied and lower representation contributed to lower sensitivity in some cases.

Table 1 - Performance of categorisation: The categorisation system's accuracy was tested against a manually classified sample of 341 actions, using a confusion matrix to calculate sensitivity, specificity and F1 scores for the overall results and for each individual category. The categorisation achieved good sensitivity (correctly identifying relevant actions) and specificity (correctly excluding irrelevant actions) across most categories.

Alt text - A table lists the categories and the computed sensitivity, specificity and F1 scores for each, with a micro-averages row at the top.

Categories	Number of actions	% of total	Sensitivity (Recall)	Specificity (Precision)	F1 score
Macro-averages - all categories	3417	100	0.97	0.78	0.85
Ongoing Visibility & Communication	490	14.3	0.71	0.90	0.79
Networking and Presenting	337	9.9	0.61	0.92	0.74
Monitoring & Evaluation of TC	315	9.2	0.58	0.92	0.71
Career Pathways & Progression	309	9.0	0.85	0.97	0.90
Apprenticeships, Internships, Placements	219	6.4	0.94	0.98	0.96
Training & Skills Development	183	5.4	0.68	0.91	0.78
External Collaboration & Partnerships	169	4.9	0.50	0.96	0.66
Recognition & Awards	162	4.7	0.84	0.98	0.90
Mentorship & Support	141	4.1	1.00	0.97	0.98
Career Frameworks & Role Definition	131	3.8	0.74	0.98	0.84
Representation in Institutional Governance	126	3.7	0.76	0.95	0.85
Technician Leadership	105	3.1	1.00	0.95	0.98
Professional Registration & Accreditation	99	2.9	0.93	0.99	0.96
Data & Workforce Analysis	82	2.4	1.00	0.98	0.99
Publications	76	2.2	0.90	0.99	0.94
EDI (Equality, Diversity & Inclusion)	75	2.2	0.56	1.00	0.71
Funding for Technicians	75	2.2	0.58	1.00	0.74
Technician Voice & Feedback	66	1.9	0.83	0.98	0.90
Teaching	64	1.9	0.83	0.95	0.89
Recruitment & Onboarding	60	1.8	0.63	0.98	0.76
Research Funding applications	45	1.3	0.86	0.99	0.92
Operational Sustainability	29	0.8	0.67	0.98	0.79
Research Assessments	19	0.6	0.85	0.99	0.91
Environmental Sustainability	17	0.5	0.73	1.00	0.84
Equipment and Asset Sharing	17	0.5	0.70	1.00	0.82

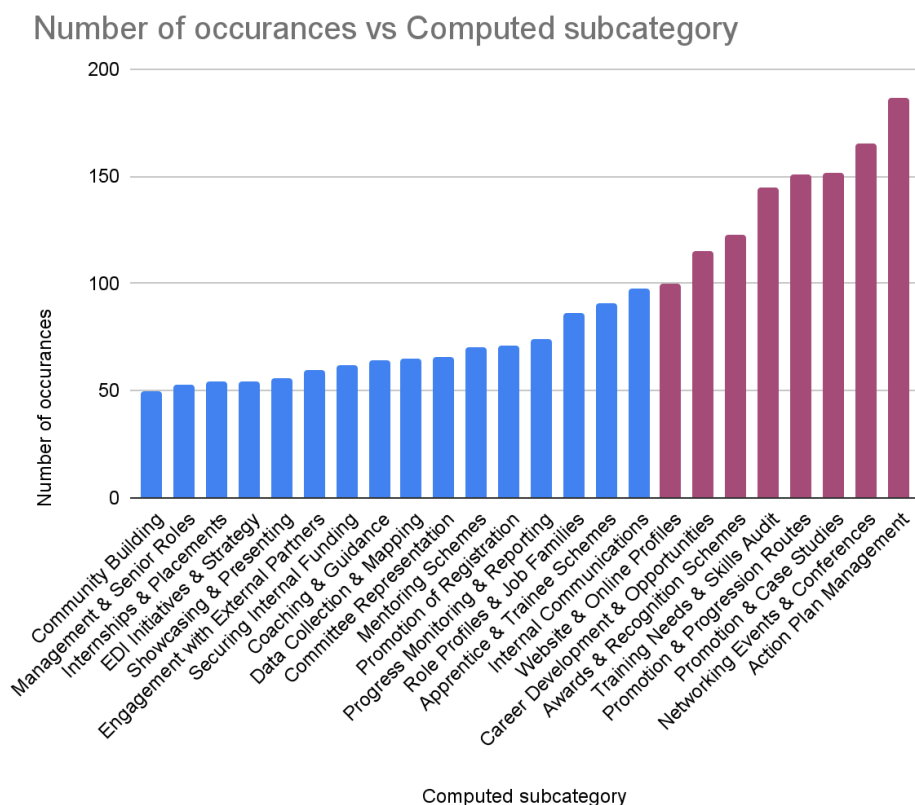
Wellbeing & Mental Health	6	0.2	0.89	1.00	0.94
Grand Total	3417				

d. Sub-themes within action categories

Automated suggestion of and categorisation into Sub-themes produced a list of more focused themes present in the data (Fig. 3; Table S3). These are subsets of their parent categories and represent more niche areas. Computation of these categories was needed to allow more specific attribution of resources which may address this theme in the next stage of the analysis. It also highlights a list of focused areas of interest indicated in the reports which was obscured by the more coarse-grained categorisation.

Figure 3 - Number of occurrences in computed subcategories: TF-IDF cosine similarity was used to search the categories for a subset of keywords which were associated with a particular niche within the category, assigned by Google Gemini. These were listed in order of number of occurrences, and a cut off of 50 applied.

Alt text - a bar graph shows the top 24 computed subcategories, with the top 8 highlighted as these were given further attention in the next section.



e. Identification and back-checking of resources related to subcategories

Systematic web searching produced a list of resources which were manually filtered to retain resources supportive to an aspect of technical staff career development. These represent potential sources of support for technicians external to their place of work, and as such could be expected to fill gaps in or align with work towards TC goals. TC Action Plans were checked against the list to identify mentions of resources within the actions, which would indicate alignment between the two.

In order to assess alignment between the Technician Commitment Action Plans and the existing resources and organisations identified, the Action Plans were searched for all resources in the list. The previously extracted 3417 actions were searched using the list of resources in Table 2 as keywords and counting all mentions in the actions, as shown in Figure 4. There were 92 mentions of external resources in 3417 actions.

Table 2 - High-level overview of support and resources identified for top 8

subcategories: In order to identify supporting organisations for which to back-check action plans, the below list was produced using a semi-automated search method. The list demonstrates that a wide range of organisations are engaged in the space, and there are already some non-specialist digital resources which can be used to promote technicians.

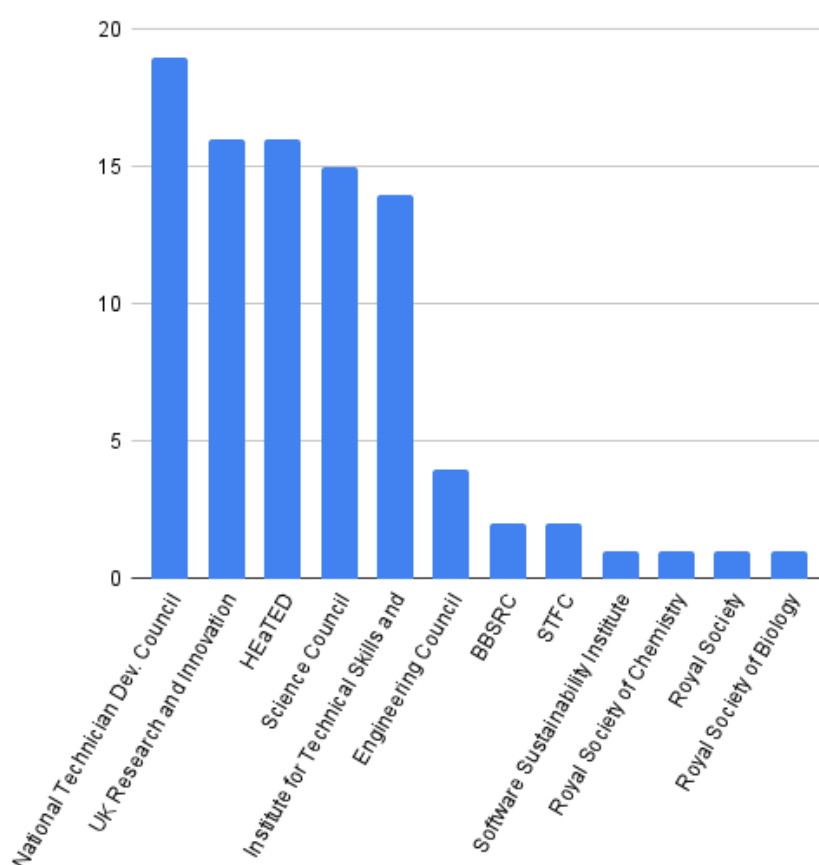
Alt text: A table showing the results of web searching to identify resources which can be used to impact technical staff careers, with a long list of organisation names and websites.

Subcategory	Score	Identified relevant resources
Website & Online Profiles	100	LinkedIn, ResearchGate, GitHub, Academia.edu, Mendeley, X, Twitter, Google Scholar
Promotion & Case Studies	152	
Career Development & Opportunities	115	General: UK Institute for Technical Skills and Strategy, Higher Education and Technicians' Educational Development (HEaTED), National Technician Development Centre (NTDC), Gatsby Foundation, UK Research and Innovation (UKRI), Science Council, The Royal Society, The Institute of Science and Technology (IST), Science and Technology Facilities Council (STFC), Dstl (Defence Science and Technology Laboratory), Skills England. Engineering: Royal Academy of Engineering, Institution of Engineering and Technology (IET), Institution of Mechanical Engineers (IMechE), Institution of Civil Engineers (ICE), Royal Aeronautical Society (RAeS), Engineering and Physical Sciences Research Council (EPSRC), Engineering Council, The TWI (The Welding Institute), The Institution of Structural Engineers (IStructE), The Energy Institute (EI), The Institute of Marine Engineering, Science & Technology (IMarEST), The Nuclear Institute, Women's Engineering Society (WES), EngineeringUK. Chemistry: Institution of Chemical Engineers (IChemE), Royal Society of Chemistry (RSC), The Society of Chemical Industry (SCI) IT, Software, Data Science: BCS, The Chartered Institute for IT, TechUK, The Chartered Institute of Information Security (CIIISec), (ISC)², NetUK, The Alan Turing Institute, Royal Statistical Society (RSS), Health Data Research UK (HDR UK), The Faculty of Clinical Informatics (FCI), The Data Lab, Centre for Data Ethics and Innovation (CDEI), ELIXIR-UK, Software Sustainability Institute Physics/Materials: Institute of Physics (IOP), Institute of Materials, Minerals and Mining (IOM3), Engineering and Physical Sciences Research Council (EPSRC), The National Physical Laboratory (NPL), The British Institute of Non-Destructive Testing (BINDT), The Institute of Measurement and Control (InstMC), UK Space Agency (UKSA)
Training Needs & Skills Audit	145	
Networking Events & Conferences	165	

		Physical environment: Natural Environment Research Council (NERC), British Antarctic Survey (BAS), National Oceanography Centre (NOC), British Geological Survey (BGS), The Geological Society of London Royal Astronomical Society (RAS), UK Centre for Ecology & Hydrology (UKCEH), The Royal Meteorological Society (RMetS), The Met Office, The Institute of Chartered Foresters (ICF), Royal Geographical Society (with IBG) Health/Biomedical: Medical Research Council (MRC), Biotechnology and Biological Sciences Research Council (BBSRC), The Royal Society of Biology, The Biochemical Society, Wellcome Trust, British Pharmacological Society, The Institute of Biomedical Science (IBMS), The Institute of Animal Technology (IAT), The Physiological Society, The Microbiology Society, Royal Pharmaceutical Society (RPS), General Pharmaceutical Council (GPhC), Nursing and Midwifery Council, Royal College of Nursing Information Science: CILIP: The Library and Information Association, Archives and Records Association (ARA), School Library Association (SLA), Information and Records Management Society (IRMS), BIALL (British and Irish Association of Law Librarians), Aslib (The Association for Information Management), UKeiG (UK e-information Group), The British Library, National Library of Scotland, National Library of Wales (Llyfrgell Genedlaethol Cymru), JISC, The Digital Curation Centre (DCC), The Museums Association, The National Archives (UK), Digital Preservation Coalition (DPC)
Awards & Recognition Schemes	123	The Papin Prize, Salters' National Awards for Science Technicians, Times Higher Education Awards. (Universities and societies also have award and recognition schemes.)
Promotion & Progression Routes	151	Internally managed at Signatory Establishments.
Action Plan Management	187	

Figure 4 - Back-checking for resources in Action plans: The list of identified resources and organisations was used as a list of search terms to search the TC action plans. The previously extracted actions were programmatically searched for the terms in the table, to identify where a TC action plan cites an external organisation as a supporting entity. 92 external resources were mentioned in 3417 actions

Alt text - A bar graph enumerating the 12 identified resources or organisations present in the list in Table 2 of identified resources.



f. Comparison with the TALENT Policy Commission recommendations

Recommendations were extracted from the TALENT Policy Commission report and compared with the actions. This gives information on which of the recommendations in this cornerstone publication have been pledged to be followed up in planned specific actions by TC Signatory organisations.

Figure 5 - TALENT Policy Commission recommendations vs actions heatmap: Action categories defined in this analysis were manually aligned with the most relevant recommendation from the TALENT Policy Commission report, and the actions in each assigned category was pooled to give an estimate of the extent to which Technician Commitment action plans were addressing the recommendations. R1, R9 and particularly R11 are well represented in the actions list, with R3, R4 and R8 the least well represented.

Alt text: A table showing the TALENT policy report recommendations aligned with the action categories and the number of actions in that category overall, with the third action number column colour coded to indicate the number of actions.



TALENT policy report recommendation (brief summary)	Aligned Action categories	Actions
R1: Strategic workforce planning for technical staff.	Monitoring & Evaluation of TC (n=31); Operational Sustainability (29)	344
R2: Flexible, cross-sector career pathways.	Career Frameworks & Role Definition (131)	131
R3: Improve technical workforce data and classification.	Data & Workforce Analysis (82)	82
R4: Targeted action on Equality, Diversity, and Inclusion (EDI).	EDI (Equality, Diversity & Inclusion; 75); Wellbeing & Mental Health (6)	81
R5: Fair costing and eligibility for technical staff on grants.	Research Funding applications (45); Funding for Technicians (75)	120
R6: Promote technical careers through outreach.	External Collaboration & Partnerships (169)	169
R7: Diversify entry routes into technical careers.	Apprenticeships, Internships, Placements (219)	219
R8: Involve technical experts in recruitment.	Recruitment & Onboarding (60)	60
R9: Define and publicise clear career pathways.	Career Pathways & Progression (309)	309
R10: Tailored professional development and registration.	Technician Leadership (10); Professional Registration & Accreditation (99)	204
R11: Recognise and increase visibility of technical contributions.	Ongoing Visibility & Communication (493); Networking and Presenting (37); Recognition & Awards (162); Publications (76); Teaching (64)	1129
R12: Ensure technical representation in decision-making.	Representation in Institutional Governance (126)	126
R13: Partner for skills development and knowledge sharing.	Training & Skills Development (183)	183
R14: Include technical voice in policy development.		n/a
R15: Proactive engagement from technical staff and managers.	Mentorship & Support (141); Technician Voice & Feedback (66)	207
R16: Create a national institute for technical skills.		n/a
	Research Assessments (19)	19
	Environmental Sustainability (17)	17
	Equipment and Asset Sharing (17)	17

g. Analysis of RAG Assessments

In total, three sets of documents describing action plans and subsequent RAG analyses were gathered from three institutions and used as the basis of this analysis, which is transferable to any Action Plan-RAG Analysis pair(s). The data here is shown for just one Institution due to space restrictions. For the three Institutions analysed, the scripts carried out 23,749 pairwise comparisons of 891 actions via Google Gemini which took 17 minutes at a self-constrained rate (see Supplementary methods section vii). The scripts produced two main outputs - a table containing metrics describing the progress of actions (Table 3), and a diagram visualising this data (Figure 6).

Table 3 - Metrics on progress of actions per category for one institution: This example shows the output following trajectory analysis, with the action counts per period, the trajectory codes they were assigned, and the computed % metrics. Composite Flow Codes (RAG Status + Trajectory) were assigned as follows: GC = Green -> Continued (successful continuation); GR = Green -> Related (successful refinement); GS = Green -> Stopped (successful completion & pivot); AC = Amber -> Continued (persistent challenge); AR = Amber -> Related (adaptive refinement); AS = Amber -> Stopped (stalled effort); RC = Red -> Continued (resilient persistence); RR = Red -> Related (strategic pivot from failure); RS = Red -> Stopped (acknowledged failure); UC/UR/US = Unassessed (no RAG data). RAG Status: G (Green) = Completed/Successful; A (Amber) = In progress/Ongoing; R (Red) = Delayed/Failed; U (Unknown) = No RAG assessment available. Columns: P1, P2, P3 = Total actions in each period (AP counts only); P1 -> P2, P2 -> P3 = Composite forward transitions (RAG status from RAG_AP + trajectory to next AP). Composite Metrics: Success rate: (All Green codes) / (Total with RAG) × 100%; Successful continuity: (GC+GR) / (GC+GR+GS) × 100%; Adaption score: (AR+RR) / (All Amber+Red) × 100%; Resilience score: (RC+AC) / (All Red+Amber) × 100%

Alt text - none needed

Category	P1	P1->P2	P2	P2->P3	P3	Success Rate (%)	Continued Rate (%)	Adaption score (%)	Resilience Score (%)
Technician Leadership	5	GC3 GR2	4	GR1 AC1 AR1 RR1	11	66.67	100.0	66.67	33.33
Ongoing Visibility & Communication	3	GR1 AC1 AR1	4	GC2 GR1 AR1	8	57.14	100.0	66.67	33.33

Training & Skills Development	3	GC2 AR1	6	GC3 AC1 RC1	6	62.5	100.0	33.33	66.67
Research Assessments	3	GC1 GR1 AC1	3	GR1 AR1 RC1	6	50.0	100.0	33.33	66.67
Recognition & Awards	0	—	5	AR3 RC1 RR1	3	0.0		80.0	20.0
Professional Registration & Accreditation	2	GC1 GR1	2	GC2	3	100.0	100.0		
Representation in Institutional Governance	1	AC1	4	GR2 RC1 RR1	2	40.0	100.0	33.33	66.67
Networking and Presenting	2	GR2	1	RR1	3	66.67	100.0	100.0	0.0
Teaching	2	GC2	1	AC1	3	66.67	100.0	0.0	100.0
External Collaboration & Partnerships	1	RC1	3	GC1 GR1 AR1	2	50.0	100.0	50.0	50.0
Research Funding applications	1	GC1	1	AR1	3	50.0	100.0	100.0	0.0
Apprenticeships, Internships, Placements	1	GC1	0	GR1	4	100.0	100.0		
Wellbeing & Mental Health	0	—	2	GC1 AC1	3	50.0	100.0	0.0	100.0
Data & Workforce Analysis	1	GR1	0	—	3	100.0	100.0		
Monitoring & Evaluation of TC	0	—	1	GC1	3	100.0	100.0		
EDI (Equality, Diversity & Inclusion)	2	GC1 GR1	0	—	1	100.0	100.0		
Uncategorised	1	GC1	0	—	1	100.0	100.0		
Operational Sustainability	1	GR1	0	—	1	100.0	100.0		
Mentorship & Support	0	—	1	AR1	1	0.0		100.0	0.0

Publications	0	—	1	AR1	1	0.0		100.0	0.0
Funding for Technicians	1	GC1	0	—	0	100.0	100.0		
Environmental Sustainability	0	—	1	GR1	0	100.0	100.0		
Career Pathways & Progression	0	—	0	—	1				
Recruitment & Onboarding	0	—	0	—	1				

Figure 6 - Progression trajectories for one institution: Computed graphical representation of actions per category per stage and the flow between those, including progression metrics. Actions in categories (RH column) are enumerated in each of the three phases, and computed metrics displayed alongside to give an overview of the institution's progress over time. Bar colors Green >15%; Orange 40-15%; Red <40%; Composite Metrics: Successful, completed = % Green | Successful, continued = % Green sustained (GC+GR) | Adaptability = Amber/Red adapted (AR+RR), scale 0.0-1.0; Resilience = Amber/Red persisted (AC+RC) scale 0.0-1.0.

Alt text: A diagram representing three stages of action plans plus metrics, arranged in rows on the page. Each action category is in a row, with a box to show the number of actions in that category per stage connected by lines to form trapezoid shapes, colour coded and ordered with the most numerous actions at the top. To the right of the trapezoid representation of that category are the metrics for that row represented as bars.

Category Flow Diagram with Composite RAG-Trajectory Metrics:
23 categories sorted by total actions | Box height $\propto$ action count | Ribbon darkness $\propto$ similarity



Ribbon darkness = similarity score (Dark = High -1.0, Light = Low -0.0) | Bar colors: Green $\geq 75\%$ | Orange 40-75% | Red $<40\%$
Composite Metrics: Successful completed = % Green | Successful continued = % Green | Adaptability = Amber/Red adopted (AR+RR), scale 0.0-1.0
Resilience = Amber/Red persisted (AC+RC), scale 0.0-1.0

Conclusions

Summary

- i. Analysis of 104 institutional action plans shows that UK higher education and research institutions are actively engaging in programmes to enhance the visibility, recognition, career development, and sustainability of their technical staff, in line with the Technician Commitment's four pillars.
- ii. Among 26 identified action types, institutions most frequently pledged actions related to Ongoing Visibility & Communication, Networking, and Career Pathways. Conversely, areas like Wellbeing & Mental Health, Environmental Sustainability, and Equipment Sharing received significantly less attention.
- iii. A direct comparison with the 16 TALENT Policy Commission recommendations reveals strong alignment in areas such as improving visibility (R11), diversifying entry routes (R7), and defining career paths (R9). However, there is less planned action corresponding to multiple other areas and some do not have direct correlates.
- iv. While a rich ecosystem of external supporting bodies and resources exists, these are rarely cited in the action plans, with even the most frequently mentioned appearing in fewer than 20% of institutional plans.
- v. Identification of the least-addressed themes (e.g. environmental sustainability, equipment/asset sharing, wellbeing/mental health) may indicate these areas are under supported in institutions and/or in the wider supporting ecosystem.

Interpreting the Gaps and Synergies

The challenge of definition and nomenclature

This study again highlights a fundamental challenge: the term 'technical staff' is broad and context-dependent. Although there is movement towards accepted definitions of the term 'technical staff', there is no defined set of terms with which to describe the detail of technical skills. The variability in action plans reveals this lack of a common language nationally, making sector-wide strategy and benchmarking difficult, and hindering wider discussion beyond national borders.

Further discussion around the identity of technicians, exploration of the boundaries of that identity and how that relates to better defining the group would produce sector-wide benefits. Also, identifying interoperable language through which to discuss technical careers centred on specialist skills would break down inter-institutional and international barriers and allow discussion of achievements based on a common framework of tangible outputs.

From pledges to progress

The strong focus on visibility, communication, and networking in the action plans demonstrates a sector-wide will to address the core Visibility and Recognition pillars of the Technician Commitment. However, the underrepresentation of actions in some other areas suggests these may be seen as secondary concerns or are being addressed through separate institutional strategies or specialist teams. These include wellbeing and mental health, research assessments, asset sharing, operational sustainability and environmental sustainability.

Wellbeing and Mental Health: Supporting Higher Education and Research Institution staff wellbeing and mental health can have positive improvements on productivity (McNeil, 2021; Nowrouzi-Kia et al., 2022; Wilkinson, 2023), but specific data on technical staff wellbeing (e.g. from institutional survey results) are not made available. Funding the collection and analysis of primary community-derived data would increase the available information in this area. Professional bodies including some of those cited in Table 2 have been active in providing wellbeing support, which would not have been captured by analysing the Higher Education and Research Institution-produced Action Plans. Linking existing programmes and expertise more explicitly into the technical staff arena would benefit both employees and employers.

Research Assessments: The role of technicians in research assessments is a developing area, for example the REF2029 criteria for submissions (Vere, 2025). The underrepresentation of this in Action Plans may indicate a lack of awareness exacerbated by the relatively undefined nature of non-research staff involvement in the exercise to this point; as more details are disseminated, this may become a higher priority.

Asset Sharing: Providing information on equipment, instruments and assets used in specialist technical fields has the potential to link individuals across institutional and geographic boundaries and create technology-based communities (ITSS, 2023). Knowledge of use of a specialist asset can also be used as a proxy for, or element of, a description of specialist skills.

Strategic skills planning: In the context of technical careers, strategic skills planning characterises actions linked to workforce planning and forward development. A lack of initiatives in this area may indicate that detailed plans of this type have not been formulated at all or are not suitable for dissemination or they are handled by other institution functions/staff.

Environmental Sustainability: Technical staff are uniquely placed to have an impact on environmental sustainability practices within research environments (Farley, 2024; ITSS, 2025c), as they are often central to running facilities, equipment, instruments or processes. The absence of environmental sustainability initiatives in the documents may indicate that environmental reforms were too embryonic at the time the plan was formulated or are being driven by other staff groups within higher education and research institutions.

An untapped ecosystem?

The significant gap between the wealth of available national resources (e.g. national and professional bodies, funding councils) and their low citation in action plans is an interesting finding. This disconnect suggests that institutions may be either unaware of the external support available or are prioritising internally-developed solutions. This exposes a crucial opportunity for such bodies to further engage with stakeholders and increase interactions directly with those drafting institutional policy.

Skills as a 'Research Currency'

What are 'Research Currencies'?

Publishing of work has been described previously as a currency of academic research, leading as it can to reputational and tangible gains such as recognition and grant funding (Hyland, 2023; Trueblood et al., 2024). Here, the term 'Research Currencies' has been broadened to mean the tangible evidence of knowledge, skills, experience and expertise built up through engaging in a career in research; 'Research Currencies' are 'earned' through carrying out successful and/or impactful work, can be 'banked' on a CV or a portfolio, and can be used to evidence achievements to peers, and in career development opportunities. In a typical academic research career, currencies which hold weight would include publications and papers, talks and presentations, and winning research funding grants. While these things are important in-and-of themselves to a researcher's work, they are also the summary evidence of sustained and significant ability in the area of focus, and of recognition within the wider research community.

'Research Currencies' and Technical Staff

Access to 'Research Currencies' for people working in technical roles varies between institution and laboratory, and often at the discretion of an individual Principle Investigator (PI) or grant holder. Some UK research funders (including UKRI and the Wellcome Trust) allow for technical staff to be principal or co-investigators on grant applications, and most publishers (including Springer-Nature and Elsevier) base inclusion as a co-author on contribution not job title, demonstrating that the opportunity for technicians to take lead roles is slowly increasing. Reflecting this, many research-intensive UK higher education and research institutions have policies or guidance around fair attribution of authorship including technical staff.

However, evidence shows that these opportunities and guidelines are often not put into practice - in one survey just 23% of technicians agreed with the statement 'Technical staff are usually credited appropriately for their contribution to research and/or research outputs', with 33% stating that PIs 'usually include them in research grant applications' (Vere and Harris, 2022). In order to bolster the credibility of individuals and the profession within the research ecosystem, it is important to increase access to existing research currencies for technical staff while retaining current quality control and publishing rigor.

In addition, sustainable technical careers will require their own ‘technical staff-specific’ measures of success and mechanisms of benchmarking against peers. This area has thus far been neglected at a national level; we propose that ‘technical skills’ or applications/demonstrations of those skills are one key base ‘currency’ of technical staff work through which benchmarking can be anchored.

Skills-based currencies

The creation of targeted mechanisms which allow staff to build up demonstrable evidence of development and use of technical skills, and how they have contributed to achievements in the workplace, would be beneficial for staff and institutions. This could be operationalised by creating mechanisms to allow individuals to curate an online ‘Skills Portfolio’ and to certify technical and specialist skills held on the Portfolio in a verifiable manner. The Portfolio would become a common home displaying and linking to multiple types of evidence, including formal training certifications, experience and skills gained through professional registration and details of projects, papers, preprints and datasets to which they have contributed.

Portfolios are already a long-established mechanism to demonstrate visual design or artistic skills and achievements; paired with a robust review and endorsement process for technical skills, this would allow effective demonstration of the level of training, ability and development of an individual. It would provide a self-maintained, granular, flexible and expedited way to demonstrate technical skills, maximising the value of learning and increasing the visibility of formal training and certification.

To underpin this, the creation and adoption of a standardised and periodically updated ‘living skills ontology’ that defines transferable skills would facilitate national and international equivalence (Robinson and Dominguez Reig, 2020).

Implications of this research for policy and practice

For higher education and research institutions

These results confirm that institutions are carrying out initiatives that map to the aspirations of the Technician Commitment. Organisations can use this evidence to critically evaluate the breadth of their action plans, ensuring they address the full spectrum of TALENT Policy Commission recommendations, including contributing to systemic issues like strategic skills/workforce planning and staff wellbeing.

The data indicates an opportunity for institutions to seek to integrate and leverage the existing national support infrastructure to avoid duplicating effort and enhance their provision. This can be supported by increasing the roles for technical staff in Institutional governance processes and with the explicit inclusion of technical staff in the outputs of evaluation exercises such as REF2029 (Vere, 2025).

For policy makers

The findings suggest a need for more prescriptive guidance to ensure a baseline of consistency across action plans, particularly regarding data collection and reporting. Robust application of progress reporting based on a digital platform linked to HESA data and including specific dates would improve the visibility of ongoing work, including the Technician Commitment; this would also allow the increased accuracy of machine learning-based tools as used in this analysis.

National bodies including Skills England may also be in a position to provide leadership around the development of a shared skills ontology to create a common language for technical roles across the sector, promoting lifelong learning and equality of opportunity by making career pathways more transparent and portable.

For technical staff and professional bodies

Technical staff can use these findings to advocate for support in underrepresented areas like publication equity, leadership training and mental health. Professional bodies can evaluate potential disconnects between their resources and institutional plans, possibly warranting more targeted outreach and partnership building with higher education and research institutions. Professional Registration can form a foundation stone of the digital portfolio mechanism, and in turn be bolstered by alignment with other achievements.

For other Higher Education processes and charters

Tracking and evidencing progress from written narrative submissions from hundreds of organisations is time consuming but necessary for reporting and audits. This work demonstrates a route by which NLP can be employed to give highly granular semi-quantitative analysis of multiple disparate charter-derived documents, which may be applicable in other areas. This in turn provides a mechanism to share practices, learn what works and what doesn't and apply this to refining best practice.

Limitations and Future Research

Limitations of this study:

- This study was undertaken by an independent researcher with extensive experience in technical roles within UK research institutions, bringing insider knowledge of technical career challenges while maintaining analytical objectivity through automated categorisation methods and systematic performance evaluation against manually validated ground truth data.
- The primary limitation of this study is its reliance on action plans, which represent intent rather than proven action. This could be remedied by a repeated analysis as more institutions progress further into the scheme and complete RAG progress reports of their pledged Actions.

- Reports from all three stages with RAG assessments were only available online for three institutions, so a sector wide analysis of progress was not possible.
- Only the most recent Action Plan for a given institution was analysed, meaning any prior completed actions were not taken into account.
- The NLP categorisation achieved good overall performance (macro-averaged sensitivity 0.97, specificity 0.78, F1 0.85) but showed variable sensitivity across categories (range 0.50-1.00). Lower-performing categories often involved generalised language (e.g., 'External Collaboration & Partnerships' sensitivity 0.50) or conceptual overlap with other categories. Categories with fewer training examples also showed reduced sensitivity. High specificity across most categories (≥ 0.90) provides confidence that assigned categories represent genuine matches. These limitations may result in underestimation of some themes, partially offset by the multi-label approach allowing relevant actions to appear in multiple categories.
- The resource identification step was based on a pragmatic but non-exhaustive search methodology, and it is likely that some resources are not included here.
- Analysis does not take into account the differing numbers of actions put forward in each plan, which will skew results towards the opinions of those who have included more actions on a given topic.
- Mapping of Action Categories to TALENT Policy Commission recommendations was carried out manually, and as such may be subject to operator selection bias.
- The use of proprietary AI models (Google Gemini 2.5 Pro and 2.0 Flash) means that training data composition and potential embedded biases are not transparent. Whilst deterministic configuration settings (temperature 0.0 for extraction) and systematic validation against manually categorised ground truth data mitigate some concerns, readers should be aware that model outputs may reflect characteristics of the unexamined training corpus. The high inter-method agreement between AI categorisation and human validation (macro-averaged F1 score 0.85) provides confidence in the reliability of the approach for this specific application.

Future Research Directions:

- The NLP methodology itself is highly transferable and could be applied to other higher education policy documents to track sector trends.
- Using a similar approach to map each individual organisation's progress through the Technician Commitment and align them with respect to progress or time would give a detailed picture of progress in the UK.
- Collaborative development of the proposed technical skills ontology, creating a robust framework for recognising and developing talent both within the UK and internationally, and a technical solution to demonstrate skills, will help bolster research and development sectors, technical professions and those who work within them.

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Authorship Contribution

S.Jackson carried out all work for the research and the manuscript.

Data availability Statement

All data used in this analysis is available through original sources in Supplementary Table S1 and through the Github repository for this study, which also holds all code (github.com/Sammijj/TCDOC_EXTRACT).

Declarations and conflicts of interest statements

- Conflicts of interest: None
- Ethics approval: Not required. All analysed documents were publicly available institutional policy documents published for accountability purposes. No personal data was processed.
- Consent for publication: N/A

Supplementary Methods and Tables

a. Supplementary methods

The workflow is outlined in Figure 1 and code and documentation is available at https://github.com/Sammijj/TCDOC_EXTRACT.

i. Collation of Corpus - Action Plan documents

Technician Commitment action plans and other documents produced by Universities are held on their websites (Table S1), in compliance with the TC. In addition, they are held on the Technician Commitment website (ITSS, 2025). Google Drive was used to collate the corpus. All available documents were downloaded and renamed in a consistent manner (by city and institution, and indicating whether it was an action plan document, RAG analysis, both or other type of document). Where there was more than one document of a type from an institution, these were also named by date of authorship to identify the most recent. Only the most recent Action Plan documents were taken forward into the analysis (listed in Table S1), and were copied to a separate folder. Python was used to script and carry out many of the functions of data extraction and cleaning, action categorisation and performance evaluation, as described below.

The collected action plans exhibited substantial heterogeneity in document structure and presentation format, which influenced the extraction and analysis approach:

Structural Variability

While around 40% of documents used structured tabular formats with explicit columns for actions, timelines, responsible parties, and success measures, others presented actions as bulleted or numbered lists embedded in narrative sections, and some used primarily prose-based descriptions with actions embedded within paragraphs under thematic headings

Content Granularity

Some institutions provided highly specific, discrete actions (e.g., "Create dedicated webpage showcasing technical staff achievements by Q2 2024"); Others used broader, programme-level commitments (e.g., "Enhance visibility of technical staff contributions").

Document Length and Density

The number of actions per plan ranged from 8 to 77 (mean 35, SD 17), reflecting different institutional approaches to action plan development. This variability likely reflects:

- Different interpretations of what constitutes a "discrete action" versus a programme
- Varying levels of institutional maturity in technical staff development planning
- Different document audiences (some plans appeared designed for internal operational use, others for external accountability)

Standardisation Strategy

To accommodate this heterogeneity, the Gemini-based extraction approach was designed to identify commitments regardless of format by instructing the model to extract "all discrete actions, tasks, or commitments" as complete, semantically meaningful statements. The model was prompted to:

- Preserve sentence completeness (avoiding mid-sentence splits)
- Exclude metadata such as timelines, responsible parties, and success measures
- Convert prose-embedded commitments into standalone action statements
- Maintain semantic fidelity to the original text while removing formatting artifacts

This approach ensured that actions from tables, lists, and prose were represented in a consistent format for categorisation, while acknowledging that the original context and framing of commitments varied substantially across documents. The impact of this standardisation was assessed through manual review of extracted actions (section ii) and validated through performance evaluation (section iv).

Document provenance and context

All analysed documents were authored by institutional staff teams typically comprising Equality, Diversity and Inclusion (EDI) officers or managers, technical services managers or directors, research culture or researcher development leads, human resources professionals and senior technical staff representatives or Technician Commitment champions.

Documents were produced in response to Technician Commitment requirements for signatory institutions to develop and publish action plans addressing the four pillars: Visibility, Recognition, Career Development, and Sustainability. The intended audiences for these documents included:

- Internal: institutional leadership, technical staff communities, HR departments
- External: the Technician Commitment initiative (ITSS), peer institutions, funders

Action plans represent a snapshot of institutional intentions at the time of authorship, typically covering 2-3 year planning periods. Documents analysed were those available as of 31/8/2025, representing the most recent publicly available action plans from each signatory institution. Where institutions had published multiple versions (e.g., renewal or updated plans), only the most recent version was analysed to represent current institutional commitments.

Access and Availability

All documents were obtained from either:

1. Individual institutional websites (typically hosted under research culture, technical services, or EDI pages)
2. The central Technician Commitment website (www.techniciancommitment.org.uk)

No access restrictions or paywalls were encountered during document collection. No institutions were contacted directly for documents; only publicly available materials were included to ensure the analysis reflected publicly accountable commitments.

ii. Extraction of action items from each document (document_extract2.py)

Authentication and Data Access

Initial access to the Google Drive repository was established using the OAuth 2.0 protocol. The script used the google-auth-oauthlib and google-api-python-client libraries to handle the authentication flow.

File Retrieval and Content Extraction

The script looped through all files within a specified parent folder in Google Drive. For each identified file, the MIME type was inspected to determine the appropriate text extraction procedure. For native Google Documents (application/vnd.google-apps.document), the content was extracted by exporting the file as plain text (text/plain) using the Google Drive API's export_media method. For Portable Document Format (PDF) files (application/pdf), the files were downloaded in-memory using the get_media method. The pdfplumber library was then used to parse the binary PDF data and extract the text content from each page. Files of other formats were converted to Google Doc or PDF prior to analysis.

Text Pre-processing

The raw text extracted from the documents underwent a reflowing process to correct for artificial line breaks introduced during optical character recognition (OCR) or text wrapping.

Action Item Extraction via Large Language Model

The pre-processed text from each document was submitted to Google's Gemini 2.5 Pro model (gemini-2.5-pro, accessed August 2025 via Vertex AI API) for analysis. The model was instructed with the following prompt:

> Prompt: "Extract all discrete actions, tasks, or commitments from this Technician Commitment action plan. Present each action as a complete sentence in a bulleted list. Do not include timelines, responsible parties, or success measures—only the action statements themselves. Ensure actions are semantically complete and avoid mid-sentence splits. Remove any formatting artefacts such as excessive whitespace or line breaks within sentences."

The model was configured with the following parameters:

- Temperature: 0.0 (deterministic output for consistency across documents)
- Max output tokens: 8192
- Top-p sampling: 0.95
- Top-k sampling: 40

Each document's first 50,000 characters were submitted to the model (all documents were within this limit after preprocessing). The extracted action items were parsed from the model's response, cleaned to remove any remaining formatting issues, and written to a comma-separated values (CSV) file with columns for the action text, originating institution, and document identifier.

Model Limitations

As a commercial large language model, Gemini's training data composition and potential biases are not publicly documented by Google. The model may reflect biases present in its training corpus. However, the deterministic temperature setting (0.0) ensures consistency across all extractions, and the subsequent validation against manually categorised ground truth data (section iv) provides objective assessment of extraction quality for this specific application domain.

iii. Categorisation of Action Items (predefined_categories6.py)

Following the extraction of action items, a second script was used to classify each item into predefined categories. This involved natural language preprocessing, the application of a weighted keyword lexicon, and a dynamic scoring algorithm to assign the most relevant categories.

Development of the keyword lexicon (keyword_categories_4.json)

Categories and Keywords were defined through iterative extraction and refinement of keywords using a human-in-the-loop implementation of Google gemini-2.5-pro and development versions of the predefined_categories6.py script. Briefly, starting with a manually curated list of 9 categories and 84 keywords likely to feature in the reports (Table S2), the script was run and evaluated, and the keyword list iterated based on the outcomes.

The development of the final keyword lexicon (Table S3) involved iterative cycles of automated categorisation, manual review, and refinement. Starting from the initial manually curated lexicon of 9 categories and 84 keywords (Table S2), the following process was repeated across 5 iterations:

1. Initial Categorisation Run: The predefined_categories6.py script was executed on the full dataset of 3,558 extracted actions using the current keyword lexicon.

2. Sample Review: A stratified sample of approximately 100 categorised actions was manually reviewed by the author, with particular attention to:

- Actions flagged as 'Uncategorised' that appeared to represent legitimate themes
- Actions assigned to categories that appeared semantically mismatched
- Actions addressing multiple themes where only one category was assigned
- Emerging themes not captured by existing categories

3. Keyword Refinement: Based on manual review findings:

- New keywords were added when recurring concepts were missed by existing categories
- Keyword weights were adjusted when categories showed poor discrimination (capturing too many irrelevant actions)
- New categories were created when clusters of uncategorised actions shared common themes
- Category definitions were refined to improve conceptual clarity

4. Performance Assessment: After each iteration, a subset of actions (~100) was manually categorised by the author to assess whether changes improved categorisation accuracy. Iterations continued until no substantial improvements were observed.

This process expanded the lexicon from 9 to 26 categories and from 84 to 552 keywords. The final lexicon was then applied to the complete dataset, with performance formally evaluated against a 10% random sample (n=341 actions) as described in section iv.

The iterative refinement process required approximately 5 rounds of testing, with diminishing returns observed after 5 iterations.

Decision Rules for Multi-Label Assignment: Actions were assigned to multiple categories when they scored above the assignment threshold for more than one category. This multi-label approach was adopted because many institutional commitments address multiple themes simultaneously (e.g., "Establish mentoring programmes with annual awards for outstanding mentors" addresses both 'Mentorship & Support' and 'Recognition & Awards'). Approximately 70% of actions received multiple category assignments, with the maximum being 5 categories for a single action.

Handling Edge Cases: Several types of actions required specific handling:

- Very brief actions (< 5 words) were categorised based on exact keyword matches only
- Actions containing negations (e.g., "address lack of diversity") were manually reviewed to ensure correct category assignment
- Actions describing establishment of committees or working groups were assigned to relevant thematic categories rather than solely to 'Representation in Institutional Governance'
- Actions that were highly generic (e.g., "Continue to support technical staff") were categorised based on context from surrounding actions when available, or flagged as 'Uncategorised' when context was insufficient

The resulting lexicon, shown in Table S3 was structured as a dictionary, mapping high-level categories to a set of specific keyword phrases, with each phrase assigned a numerical weight to signify its importance or relevance to the category.

Natural Language Preprocessing (predefined_categories6.py)

The input for this stage was the CSV file generated by the preceding extraction script, containing a list of discrete action items, and the keyword lexicon, which was loaded from a JSON file. To standardise both the action items and the keyword lexicon for comparison, a multi-step text preprocessing pipeline was employed using the Natural Language Toolkit (NLTK) library. For any given text string, the procedure was as follows:

- **Tokenization:** The text was converted to lowercase and segmented into individual tokens.
- **Filtering:** Non-alphanumeric tokens were removed. A standard English stopwords list was applied to remove common, low-information words, with a set of custom exceptions to retain domain-specific context.
- **Part-of-Speech (POS) Tagging:** Each remaining token was assigned a POS tag (e.g., noun, verb, adjective) using NLTK's averaged_perceptron_tagger.
- **Lemmatization:** Tokens were reduced to their dictionary base form (lemma) using the WordNetLemmatizer. The POS tag from the previous step was used to track word characteristics for the lemmatization process, ensuring, for example, that verbs and nouns were treated according to their grammatical function.

Weighted Scoring and Categorisation Algorithm

An algorithm was developed to score and categorise each action item. For a given action, its set of unique lemmatized tokens was generated. The algorithm then iterated through each category in the keyword lexicon comparing with the token set. A multi-word keyword phrase was considered a match if all of its constituent tokens formed a subset of the action item's token set. A category's score was calculated by summing the weights of all its keyword phrases that were found within the action item.

Output Generation

The final categorised data was written to a new CSV file and the assigned categories were appended using a 'one-hot' encoding scheme. This resulted in a binary column for each possible category, where a value of '1' indicated the assignment of that category to the action item in that row. An additional 'Uncategorised' column was added to flag items that did not meet the scoring threshold for any category.

iv. Performance Evaluation (evaluate_multi_col.py)

To quantitatively assess the performance of the action item categorisation algorithm, the algorithm's output was compared against a manually curated ground truth dataset, and sensitivity and specificity were calculated for each category.

Ground-truth classification

The ground truth file was prepared by manually assigning a single, correct category to each action item to 10% of the total actions (n=341). Categories containing less than 10 actions were then supplemented with additional actions by manually searching the whole actions list for strong keywords and categorising.

Data Ingestion and Alignment (evaluate_multi_col.py)

The evaluation process used two data sources: the one-hot encoded output file from the categorisation script, serving as the predictions; and a separate CSV file containing the ground truth classifications. The text in both the prediction and ground truth files was normalised by converting to lowercase and standardising whitespace, and the two datasets were then merged into a single dataframe.

Performance Metric Calculation

For each category, a confusion matrix was generated using the scikit-learn library; true labels were derived from the ground truth column, while predicted labels were taken from the corresponding one-hot encoded column in the merged dataset. From this matrix, the counts of True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) were determined.

Evaluation Metrics

Two primary metrics were calculated to evaluate the model's performance for each category:

- **Sensitivity (Recall):** The proportion of actual positives that were correctly identified, calculated as: $\text{Sensitivity} = \text{TP} / (\text{TP} + \text{FN})$
- **Specificity:** The proportion of actual negatives that were correctly identified, calculated as: $\text{Specificity} = \text{TN} / (\text{TN} + \text{FP})$

Reporting

The results of the analysis were compiled into a final summary csv file providing the sensitivity, specificity, and the raw confusion matrix counts (TP, FP, TN, FN) for each category.

Inter-rater Reliability and Validation

The final keyword lexicon and categorisation algorithm were developed through iterative refinement by a single researcher, with validation performed against a 10% random sample (n=341) of manually categorised actions. Whilst single-coder approaches are common in computational text analysis where algorithms provide consistency, the performance metrics (sensitivity, specificity, F1 scores) provide objective indicators of categorisation reliability comparable to traditional inter-rater agreement measures.

During lexicon development, a sample of categorisations (n=50 actions across 8 categories) was independently reviewed by two domain experts (senior technical services professionals from UK research institutions) to confirm face validity of category assignments. Reviewers were asked to assess whether category labels accurately represented the assigned actions and whether any major themes appeared to be missing. This review confirmed the conceptual coherence of the 26-category taxonomy and did not identify any substantial missing themes, providing qualitative validation of the categorisation framework beyond the quantitative performance metrics.

v. Development of a sub-category (“sub-themes”) classification lexicon

Prior to rules-based sub-classification of action items, the most salient keywords within each high-level category were identified. This data-driven approach was designed to provide an empirical basis for the manual definition of sub-categories.

Keyword Salience Calculation using TF-IDF (subclus_calc_weight.py)

The data was split into category-specific corpora. For each, a Term Frequency-Inverse Document Frequency (TF-IDF) analysis was conducted using the `TfidfVectorizer` from the `scikit-learn` library.

This technique was chosen to identify terms that were not only frequent within a category but also distinctive. The vectorizer was configured with sublinear term frequency scaling (`sublinear_tf=True`) to reduce the impact of very high-frequency words and used a standard English stopwords list to filter out common, non-descriptive terms.

The TF-IDF scores for each term were summed across all documents in the corpus, producing a single, cumulative importance score for every term within the context of its category.

Output for Rules-Based Sub-categorisation

The terms were ranked based on their cumulative TF-IDF scores. For each high-level category, the top 15 keywords with the highest scores were identified. These ranked lists were compiled into a text file.

A hierarchical lexicon was automatically curated by Google Gemini based on the keywords identified in the TF-IDF analysis. This lexicon was structured as a Python dictionary where each key represented a high-level category. The value for each key was a further dictionary containing a set of predefined sub-cluster names, which in turn mapped to a list of specific keywords defining the theme of that sub-cluster.

vi. Sub-Category Classification (rule_based_class.py)

A script was written to carry out this function (rule_based_class.py). For each action in the dataset, its high-level category was first identified to select the appropriate set of sub-cluster rules from the lexicon. The algorithm then applied a keyword frequency scoring method. For each potential sub-cluster, the script counted the occurrences of its associated keywords within the text of the action item.

An action item was assigned to the sub-cluster that achieved the highest keyword match count. In cases where no keywords from any sub-cluster were found, the item was labeled as 'Unassigned'.

The final output of this process was a new CSV file, with a new column containing the assigned sub-cluster name for each action item.

vii. Analysis of RAG Assessments (rag_extract_simple12.py, rag_alignment_workflow_unpaired8.py)

Document Acquisition and Multi-Format Extraction

Action Plans and RAG assessment documents were retrieved programmatically from a Google Drive repository using the Google Drive API v3 with OAuth 2.0 authentication. The workflow handled three document formats: native PDF files (extracted using pdfplumber), Microsoft Word documents (.docx files, extracted using python-docx), and Google Docs (exported to DOCX format before processing). Document metadata (institution name, document type, assessment period number) was extracted from filenames using regular expression patterns supporting multiple naming conventions.

Structured Table Extraction Algorithm

For documents containing structured action tables with explicit RAG status columns, an intelligent column detection system was implemented (rag_extract_simple12.py, lines 223-517). The algorithm:

1. Scanned table headers (rows 0 and 1) to identify columns containing "action", "step", or "what is to be done" for action text, and columns labeled "RAG" for status indicators.
2. When multiple potential action columns were detected, selected the optimal column by prioritising exact "action" header matches, then choosing the column with the most text content in the first data row.

3. Tracked column positions across continuation tables (tables lacking headers but continuing data from a previous table).
4. Merged multi-row actions where text was split across consecutive table rows, using heuristics based on capitalisation patterns and sentence completion markers.
5. Extracted RAG status using explicit markers (single letters R, A, G, or full words Red, Amber, Green), converting to standardised single-letter format.

Unstructured Text Extraction via Large Language Model

For documents lacking structured tables, action extraction was performed using Google's Gemini 2.0 Flash model (gemini-2.0-flash-001) via the Vertex AI API (rag_extract_simple12.py, lines 520-574). The model was instructed to:

1. Identify discrete actions or commitments as complete, coherent sentences.
2. Remove RAG status indicators from the action text itself (to prevent status labels contaminating semantic matching).
3. Preserve sentence completeness (avoiding mid-sentence splits).
4. Clean formatting artifacts including excess whitespace and line breaks within sentences.

The Gemini prompt included the first 50,000 characters of each document and returned actions as newline-separated text, which was post-processed to remove remaining RAG indicators using regular expressions.

Action Categorisation

All extracted actions (both from tables and unstructured text) were categorised using the same keyword taxonomy and NLTK-based lemmatisation approach described in section ii(c). The categorisation step populated the "Category" column for every extracted action, enabling category-level trajectory analysis.

Output Format

Extraction produced a CSV file (rag_actions_extraction.csv) with columns: LineID (sequential identifier), Institution (organisation name), Document Type (AP1, AP2, AP3, RAG_AP1, RAG_AP2, RAG_AP3), Action (full action text), RAG value (R, A, or G for RAG documents; empty for Action Plans), and Category (populated by categorisation algorithm).

Two-Tier Action Matching Strategy

Following extraction, actions were matched across documents and time periods using a three-tier strategy (rag_alignment_workflow_unpaired8.py):

Tier 1: Within-Period TF-IDF Matching (same assessment period)

For action pairs from the same period (e.g., AP1 vs RAG_AP1), TF-IDF vectorization (sklearn.feature_extraction.text.TfidfVectorizer) was used, followed by cosine similarity calculation. Actions with similarity ≥ 0.95 were classified as "Identical", ≥ 0.85 as "Match", and ≥ 0.70 as "Similar". This fast, deterministic method was appropriate for same-period comparisons where actions were expected to be nearly identical. Matching was performed bidirectionally: Action Plan actions were matched to their corresponding RAG assessment

actions (AP->RAG_AP), and vice versa (RAG_AP->AP). Results were stored in columns "Match AP vs RAG_AP (LineID, Status)" and "Match RAG_AP vs AP (LineID, Status)".

Tier 2: Cross-Period Batch AI Comparison (subsequent periods)

For tracking action evolution across periods (e.g., RAG_AP1 -> AP2), Gemini AI semantic analysis with batch optimisation was employed (rag_alignment_workflow_unpaired8.py, lines 289-383). This approach:

1. Compared one RAG-assessed action against up to 10 subsequent Action Plan actions in a single API call (reducing API costs ~10-fold compared to individual comparisons).
2. Included category metadata in the comparison prompt to provide semantic context (e.g., "[Categories: Training; Development]").
3. Requested the model classify relationships as:
 - "Identical": Essentially unchanged action
 - "Extended": RAG action evolved into more ambitious version
 - "Narrowed": RAG action became more focused/specific
 - "Related": Connected but different focus
 - "Unrelated": No meaningful connection
4. Returned matches as JSON with relationship type and confidence score (0.0-1.0), filtering to include only matches with score ≥ 0.5 .
5. Selected the top 5 matches by score for each source action.

AI Model Configuration and Prompt Engineering

Cross-period action matching employed Google's Gemini 2.0 Flash model (gemini-2.0-flash-001, accessed [August-September 2025] via Vertex AI API) with the following prompt template for batch comparisons:

> Prompt Template:

> ```

> Compare this action from period [N]:

>

> ACTION: "[SOURCE_ACTION_TEXT]"

> CATEGORIES: [CATEGORY_LIST]

>

> Against these actions from period [N+1]:

>

> 1. "[CANDIDATE_ACTION_1]" [Categories: ...]

> 2. "[CANDIDATE_ACTION_2]" [Categories: ...]

> [up to 10 candidates]

>

> For each candidate action, classify the relationship as:

> - "Identical": Essentially unchanged action, same focus and scope

> - "Extended": Source action evolved into more ambitious or broader version

> - "Narrowed": Source action became more focused or specific

> - "Related": Connected theme but different focus or approach

> - "Unrelated": No meaningful connection

>

```

> Return ONLY valid JSON with this structure:
> {
>   "matches": [
>     {"action_number": 1, "relationship": "Extended", "confidence": 0.85},
>     ...
>   ]
> }
>
> Include only matches with confidence  $\geq 0.5$ . Return empty array if no matches meet
threshold.
> ```

```

Model Configuration:

- Temperature: 0.2 (slight variability for nuanced semantic matching, whilst maintaining consistency)
- Max output tokens: 2048
- Response format: JSON with structured schema
- Top-p sampling: 0.95

Failure Handling

When Gemini API was unavailable (due to quota limits, network issues, or API errors), the workflow automatically reverted to TF-IDF cosine similarity matching (threshold ≥ 0.70 for "Match" classification), preserving functionality whilst sacrificing relationship-type classification granularity. All API responses were logged to a debug JSON file (`rag_actions_extraction_MATCHED_gemini_debug.json`) for quality assurance and reproducibility. In practice, API failures occurred in fewer than 1% of requests during this analysis.

Rate Limiting and API Quota Management

To avoid exceeding Vertex AI free-tier quotas, conservative rate limiting was implemented (15 requests per minute) using a sliding window approach (`rag_alignment_workflow_unpaired8.py`, lines 177-196). Request timestamps were tracked in a deque structure, with automatic sleeping when approaching the per-minute limit. Additionally, a 4-second pause was enforced between batch comparisons.

Checkpoint Architecture for Resumability

Given the computational intensity of processing hundreds of actions across multiple institutions and periods, a checkpoint system was implemented allowing workflow resumption after interruptions (`rag_alignment_workflow_unpaired8.py`, lines 94-175). After completing each major task (categorisation, within-period matching) and each institution-period transition in cross-period matching, the workflow saved:

1. Current dataframe state to CSV (`checkpoints/checkpoint_name.csv`)
2. Workflow metadata to JSON (`checkpoints/workflow_state.json`), including last completed institution, period transition, and progress metrics

On restart, the workflow detected existing checkpoints and offered resumption from the last saved state, preventing redundant API calls and computation.

Trajectory Classification

Following the three-tier matching process, each action was assigned a trajectory code:

- C (Continued): Action persists with same focus (from "Identical" or "Extended" relationships)
- R (Related): Action has related successor with different focus (from "Related" or "Narrowed" relationships)
- S (Stopped): No successor action identified
- N (New): Action with no predecessor
- U (Unknown): Insufficient data to classify

Derived Metrics for Organisational Analysis

Building on the RAG (Red-Amber-Green) status assessment and trajectory analysis, four composite metrics were developed that integrate completion status with action evolution patterns to provide a nuanced understanding of institutional progress.

Composite Code Generation: For each action in the RAG analysis periods (RAG_AP1 and RAG_AP2), composite codes were created combining RAG status (Green (G, completed), Amber (A, ongoing), or Red (R, delayed/failed)) with Trajectory type (Continued (C), Related (R), or Stopped (S)). These were derived by matching each RAG-assessed action to its corresponding action in the subsequent action plan using the Gemini AI fuzzy matching described above. For example, a "Green" action that was "Extended" in the next period receives the composite code "GC" (Green-Continued), while an "Amber" action that was "Narrowed" receives "AR" (Amber-Related).

Trajectory Classification: Gemini relationship types were classified into three trajectory categories:

- Continued (C): Actions classified as "Identical", "Extended", or "Continued", Indicating the action was sustained with similar or expanded scope
- Related (R): Actions classified as "Related", "Narrowed", "Partial", or "Replaced", Indicating the action evolved, was refined, or shifted focus
- Stopped (S): Actions classified as "Stopped" or "Unrelated" Indicating the action was discontinued or replaced with unrelated work

Composite Metrics: Four complementary metrics were calculated for each action category, aggregating composite codes across both transition periods (RAG_AP1→AP2 and RAG_AP2→AP3):

1. Successful, Completed (%) - Percentage of RAG-assessed actions that achieved completion (Green status), regardless of whether they continued or stopped.
Formula: $(\text{All Green codes} / \text{Total actions with RAG status}) \times 100$

Interpretation: Measures overall success in completing committed actions. High values indicate strong delivery on commitments. This metric captures both actions completed and sustained (GC, GR) and those completed but discontinued (GS).

2. Successful, Continued (%) - Among successfully completed actions (Green), the percentage that were sustained or evolved (GC + GR) rather than stopped (GS).
Formula: $((GC + GR) / (GC + GR + GS)) \times 100$
Interpretation: Measures whether successfully completed actions are maintained in subsequent planning cycles. High values suggest completed work is embedded into ongoing practice rather than treated as one-off achievements. Low values may indicate completion of temporary initiatives or shifting strategic priorities.
3. Adaptability Score (0.0-1.0) - Among challenged actions (Amber or Red status), the proportion that were adapted or refined (AR + RR) rather than continued unchanged or stopped. Formula: $((AR + RR) / (All\ Amber + All\ Red)) \div 100$

Interpretation: Measures the institution's capacity to learn from difficulties and adjust struggling actions. High scores indicate responsive, iterative improvement. Low scores may suggest either abandonment of difficult work or persistence with unchanged approaches despite challenges.

4. Resilience Score (0.0-1.0) - Among challenged actions (Amber or Red status), the proportion that were sustained unchanged (AC + RC) despite difficulties. Formula: $((AC + RC) / (All\ Amber + All\ Red)) \div 100$

Interpretation: Measures institutional commitment to maintaining difficult actions. High scores indicate persistence and determination. When considered alongside Adaptability Score, provides insight into institutional response to challenges: high resilience with low adaptability suggests rigid persistence, while low resilience with high adaptability suggests responsive refinement.

Metric Interpretation: These metrics are designed to be interpreted together rather than in isolation:

- High Success + High Successful Continuity: Evidence-based approach with sustainable embedding of completed work (e.g., strong institutional capacity)
- High Success + Low Successful Continuity: Completion-focused approach with frequent strategic pivots (e.g., responsive to changing priorities)
- Low Success + High Adaptability: Learning-oriented approach actively refining struggling actions (e.g., developmental phase)
- Low Success + High Resilience: Committed approach persisting despite challenges (e.g., addressing systemic barriers)
- Low Success + Low Adaptability + Low Resilience: Indicates discontinued work, potentially reflecting resource constraints or changing institutional priorities

The combination of these metrics across action categories enables identification of institutional "learning profiles" - characteristic patterns in how organisations approach, sustain, and evolve their equity, diversity, and inclusion commitments over time.

Fallback Strategy and Robustness

When Gemini AI was unavailable (due to quota limits, network issues, or API errors), the workflow automatically fell back to TF-IDF similarity for cross-period matching, preserving

functionality at the cost of reduced relationship-type specificity. All API responses from Gemini were cached to a debug JSON file (rag_actions_extraction_MATCHED_gemini_debug.json) for quality assurance and reproducibility. In practice this was not used in this analysis.

Final Output

The workflow produced an enriched CSV file (rag_actions_extraction_MATCHED.csv) containing:

- All original extraction columns (LineID, Institution, Document Type, Action, RAG value, Category)
- Within-period match columns showing bidirectional AP<->RAG_AP correspondences
- Cross-period match columns showing RAG_AP(n) -> AP(n+1) trajectories with relationship classifications and scores
- Trajectory codes and derived metrics enabling analysis of institutional commitment evolution over time

This comprehensive approach enabled quantitative analysis of how institutional commitments evolved, which action types were sustained or discontinued, and which categories showed dynamic growth versus stability.

Visualisation

To enable intuitive interpretation of category-level action evolution across three assessment periods, a graphical summary output was developed using matplotlib (create_graphical_summary_multi_Institution.py) alongside an output table of the key computed metrics (create_comprehensive_summary_multi_Institution.py). In the category flow diagrams, these metrics are displayed as horizontal bars to the right of the period columns.

viii. Identification of resources available to the technical staff community

The aim of this part of the analysis was to identify the additional support “Technical staff” can expect to be able to call on outside of their University or employer’s commitments, or between posts. The ten top-scoring sub-categories identified by the analysis in (f) above were used as exemplars to examine the provision of support and resources in each area outside of the signatory Universities.

To enable a wide search and identify any applicable websites or resources, information on the status of each action area was searched for manually using Google Search. The following search criteria were applied, with the ‘Verbatim’ setting switch to ‘On’, searching from the United Kingdom region:

[subcategory title] AND (UK OR united kingdom) AND (technical staff OR technician OR technologist)

The Google Chrome plugin ‘SERP Sonar’ was used to extract the search results, page titles and summaries to a csv file. The first 50 results for each were examined and relevant information was tabulated. Resources were added to a table and supplemented with other

known resources. This allowed identification of support for provision compared to the ambitions of the Action Plan documents, and the identification of gaps in this support.

ix. Back-checking of presence of identified resources in Actions

A script was written to carry out this function (`enumerate_resources.py`). A lexicon of resource-related keywords was formed to represent the specific entities included in the tables following identification in (g) above. This new dictionary was incorporated into a JSON and applied again to the original extracted list of actions using a python script (`enumerate_resources.py`), with the aim of enumerating instances where these key resources had been mentioned in the action plans. Mentions were screened for instances of self-referral and remaining mentions put into a table. This was designed to provide a measure of the concordance between planned actions and existing resources, and examine the extent to which existing infrastructure or resources were being cited in action planning.

x. Comparison of TALENT Policy Commission report recommendations with actions and resources

The TALENT commission report includes general and sector specific recommendations for implementation of its findings. The 16 recommendations were extracted from the text and manually aligned with the 26 categories resulting from the Action Plan documents. This identified the level of discussion of information pertinent to each recommendation in the Action Plans.

Table S1 - List of action plans, institution, links

Organisation	Action Plan source
Anglia Ruskin University	https://www.techniciancommitment.org.uk/assets/action-plans-renewal/appendix-1-stage-2-ap.pdf
Aston University	https://www.aston.ac.uk/about/mission/technician-commitment
Babraham Institute	https://www.techniciancommitment.org.uk/assets/action-plans-renewal/babraham-institute-july-2024.pdf
Birmingham City University	https://bcuassets.blob.core.windows.net/docs/csu2024471-technicians-commitment-brochure4v10int29-1-25-133827190577162904.pdf
British Geological Survey	https://www.techniciancommitment.org.uk/assets/annex-a-bgs-technician-commitment-action-plan.docx
Brunel University of London	https://www.brunel.ac.uk/about/documents/pdf/Action-Plan.pdf
Buckinghamshire New University	https://www.techniciancommitment.org.uk/assets/stage-1-buckinghamshire-tc-plan-final.pdf
Canterbury Christ Church University	https://www.techniciancommitment.org.uk/assets/action-plans-renewal/canterbury-action-plan.pdf
Cardiff Metropolitan University	https://www.cardiffmet.ac.uk/about/equality-and-diversity/technician-commitment/
Cardiff University	https://www.cardiff.ac.uk/documents/2915574-technician-commitment-action-plan-20242027
Coventry University	https://www.techniciancommitment.org.uk/assets/action-plans-renewal/coventry-university-tc-action-plan-2023-2025.pdf
Cranfield University	https://www.cranfield.ac.uk/about/working-at-cranfield/technician-commitment
De Montford University	https://www.dmu.ac.uk/technicians/action-plan-self-assessment.aspx
Durham University	https://www.dur.ac.uk/about-us/technician-commitment/action-plan/
Earlham Institute	https://www.earlham.ac.uk/sites/default/files/2023-12/EI%20Technician%20Commitment%20Action%20Plan%2030112023.pdf
Falmouth University	https://www.techniciancommitment.org.uk/assets/falmouth-university-technician-commitment-action-plan.pdf
Francis Crick Institute	https://www.crick.ac.uk/careers-study/scientific-staff/technician-commitment-action-plan-2021-2024#:~:text=The%20Crick%20is%20a%20signatory.by%20technical%20and%20scientific%20staff.
Harper Adams University	https://cdn.harper-adams.ac.uk/document/page/422_Technician-Commitment-Action-Plan.pdf
Health Data Research UK	https://www.hdruk.ac.uk/wp-content/uploads/2023/06/HDR-UK-Technician-Commitment-Action-Plan_2023-1.pdf
Heriot Watt University	https://www.hw.ac.uk/documents/technician-commitment-action-plan-hwu.pdf
Imperial College London	https://www.imperial.ac.uk/technicians-portal/technician-commitment/technician-commitment-action-plan/
Institute of Cancer Research	https://www.techniciancommitment.org.uk/assets/action-plans-renewal/appendix-3.pdf
James Hutton Institute	https://www.techniciancommitment.org.uk/assets/action-plans/james-hutton-institute-action-plan-and-self-assessment-final.pdf
John Innes Centre	https://www.jic.ac.uk/app/uploads/2024/01/TC-Appendix-3-Future-Action-Plan-2024-to-2026_final.pdf

Keele University	https://www.keele.ac.uk/organisationaldevelopment/techniciancommitment/
Kings College London	https://www.kcl.ac.uk/research/research-environment/developing-research-proposals/technician-commitment#section-3
Lancaster University	https://www.techniciancommitment.org.uk/assets/action-plans-renewal/lancaster-action-plan.pdf
Liverpool John Moores University	https://www.techniciancommitment.org.uk/assets/action-plans/ljmu-technician-commitment-initial-action-plan.pdf
Liverpool School of Tropical Medicine	https://www.lstmed.ac.uk/research/lstm-technician-commitment
London South Bank University	https://www.lsbu.ac.uk/about-us/technician-commitment
Loughborough University	https://www.lboro.ac.uk/services/od-hub/topics/technician-support/
Manchester Metropolitan University	https://www.mmu.ac.uk/about-us/professional-services/academic-services/technical-services/technician-commitment
MRC Harwell Institute	https://www.har.mrc.ac.uk/about-us/jobs/technician-commitment/
MRC Laboratory of Medical Science	https://www.techniciancommitment.org.uk/assets/action-plans-renewal/lms-technician-commitment-3-year-action-plan-final-26-sept-2023-new-logo.pdf
MRC Laboratory of Molecular Biology	https://www.techniciancommitment.org.uk/assets/action-plans/lmb.pdf
Newcastle University	https://www.ncl.ac.uk/research/technicians-community/
Northumbria University	https://www.techniciancommitment.org.uk/assets/action-plans/final-northumbria-university-technician-commitment-action-plan-2022.pdf
Nottingham Trent University	https://www.ntu.ac.uk/media/documents/jobs/technician-commitment-36month-action-plan.pdf
Open University	https://stem.open.ac.uk/sites/stem.open.ac.uk/files/files/3b_Appendix 1_OU_ActionPlan_2024-27.pdf
Oxford Brookes University	https://www.techniciancommitment.org.uk/assets/action-plans/technician-commitment-action-plan-oxford-brookes.pdf
Queen Mary University of London	https://www.qmul.ac.uk/technician-commitment/
Queens University Belfast	https://www.qub.ac.uk/sites/technicians/TechnicianCommitment/
Royal Holloway University	https://www.royalholloway.ac.uk/research-and-education/technicians/technician-commitment/
Science and Technology Facilities Council	https://www.technicians.org.uk/assets/technician-commitment/pdfs/stfc-technician-commitment-action-plan.pdf
Sheffield Hallam University	https://www.technicians.org.uk/assets/technician-commitment/pdfs/sheffield-hallam-university-technician-commitment-action-plan-24-05-19.pdf
St Andrews University	https://www.st-andrews.ac.uk/osds/my-development/professional-staff/technicians-commitment-initiative/
Swansea University	https://www.techniciancommitment.org.uk/assets/action-plans/tc-swansea-action-plan-swansea-23-24.pdf
The Sainsbury Laboratory	https://www.tsl.ac.uk/working-at-tsl/support-for-researchers/technicians-commitment
UK Research and Innovation	https://www.ukri.org/wp-content/uploads/2021/02/UKRI-040221-Tech-nicianCommitmentActionPlan.pdf

Ulster University	https://www.techniciancommitment.org.uk/assets/action-plans/uu-action-plan-tc2023.pdf
University College London	https://www.ucl.ac.uk/human-resources/organisational-development/technical-staff-development
University for the Creative Arts	https://www.uca.ac.uk/about-us/technical-services-commitment/
University of Aberdeen	https://www.abdn.ac.uk/staffnet/working-here/edi/technician-commitment/
University of Bath	https://www.bath.ac.uk/corporate-information/technician-commitment/
University of Birmingham	https://www.birmingham.ac.uk/documents/college-les/technician-commitment-stage-3-self-assessment-action-plan-uob-final-signed-accessible-2024.docx
University of Bradford	https://www.bradford.ac.uk/about/our-structure-and-organisation/our-technicians/technician-commitment/Technician-Commitment---Action-Plan.pdf
University of Brighton	https://www.techniciancommitment.org.uk/assets/action-plans/uni-of-brighton-action-plan-jan-2021-jan-2023.pdf
University of Bristol	https://www.bristol.ac.uk/media-library/sites/staffdevelopment/documents/pssd/technician-commitment-stage-3-self-assessment-action-plan-guidance%20(final)%20020724.pdf
University of Cambridge	https://www.technicians.admin.cam.ac.uk/technician-commitment/action-plan
University of Chester	https://www.chester.ac.uk/about/jobs/technician-commitment-initiative/
University of Derby	https://www.derby.ac.uk/services/technical-services/our-technician-commitment/
University of Dundee	https://www.dundee.ac.uk/technicians
University of East Anglia	https://www.techniciancommitment.org.uk/assets/action-plans/2021-02-28-uea-action-plan-submission-version.pdf
University of Edinburgh	https://technicians.ed.ac.uk/sites/default/files/2024-11/UoE_TC%20Action%20Plan_2024-2026_web.pdf
University of Essex	https://www.essex.ac.uk/information/equality-and-diversity/technician-commitment#:~:text=The%20Commitment%20aims%20to%20ensure.pdf
University of Exeter	https://www.exeter.ac.uk/departments/tso/staffdevelopment/techniciancommitment/
University of Glasgow	https://www.gla.ac.uk/myglasgow/staff/technicians/futureactions/
University of Greenwich	https://www.techniciancommitment.org.uk/assets/action-plans/university-of-greenwich-technician-commitment-action-plan.pdf
University of Hertfordshire	https://www.herts.ac.uk/staff/technical-staff
University of Huddersfield	https://www.hud.ac.uk/media/assets/document/hudtechnet/TC-Stage2ActionPlan.pdf
University of Hull	https://www.techniciancommitment.org.uk/assets/action-plans/technician-commitment-university-of-hull-2020.pdf
University of Kent	https://media.www.kent.ac.uk/se/48555/KentTechnicianCommitmentsStrategyandActionPlan_2024-2027.pdf
University of Leeds	https://researchculture.leeds.ac.uk/technician-commitment/
University of Leicester	https://le.ac.uk/the-technician-commitment/plan
University of Lincoln	https://techniciansmakeithappen.blogs.lincoln.ac.uk/2022/03/21/technicians-commitment-year-3-action-plan/

University of Liverpool	<u>https://www.liverpool.ac.uk/media/livacuk/researcher/techhub/2024-27-action-plan.pdf</u>
University of Manchester	<u>https://www.manchester.ac.uk/research/environment/technical-support/commitment/#:~:text=The%20University%20of%20Manchester%20is,the%20institutions%20they%20critically%20support</u> .
University of Northampton	<u>https://www.techniciancommitment.org.uk/assets/action-plans/uon-technician-commitment-action-plan-phase-1-submission.pdf</u>
University of Nottingham	<u>https://www.nottingham.ac.uk/technicians/documents/uon-technician-commitment-action-plan-2024.pdf</u>
University of Oxford	<u>https://www.technicians.org.uk/assets/technician-commitment/pdfs/university-of-oxford-technician-commitment-action-plan-2.pdf</u>
University of Plymouth	<u>https://www.plymouth.ac.uk/rails/active_storage/blobs/proxy/eyJfcmlkLWVzZWZlIjBHaHBBN05VRVE9PSIsImV4cCI6bnVsYmFvcyJmXvYmxvYl9pZCJ9fQ==--c2f449eca1bbf934763589bc02985572039bb3b0/UoP_Technician_Commitment_Action_Plan_2024-27.pdf.pdf</u>
University of Portsmouth	<u>https://www.port.ac.uk/about-us/working-at-portsmouth/technician-commitment</u>
University of Reading	<u>https://www.reading.ac.uk/technical-services/more-about-technical-services</u>
University of Salford	<u>https://www.salford.ac.uk/technicians/technician-commitment</u>
University of Sheffield	<u>https://www.sheffield.ac.uk/technicians/championing#:~:text=Our%20University%20Technician%20Commitment%20Action,%2C%20Career%20Development%2C%20and%20Sustainability</u> .
University of Southampton	<u>https://www.southampton.ac.uk/technicians/technician-commitment/index.page</u>
University of Staffordshire	<u>https://www.staffs.ac.uk/about/corporate-information/technician-commitment</u>
University of Strathclyde	<u>https://www.strath.ac.uk/media/1newwebsite/humanresourcesvacancies/downloads/Technician_Commitment_Plan_2024-2027.pdf</u>
University of Sussex	<u>https://www.sussex.ac.uk/staff/technicians/technical-commitment#:~:text=The%20University%20of%20Sussex%20has,Keith%20Jones</u> .
University of the West of England	<u>https://www.uwe.ac.uk/about/values-vision-strategy/equality-diversity-and-inclusivity/technician-commitment</u>
University of Warwick	<a href="https://warwick.ac.uk/research/technicians/ourtechniciancommitment/<u>https://warwick.ac.uk/research/technicians/ourtechniciancommitment/</u></td></tr> <tr> <td>University of York</td><td> <u>https://www.york.ac.uk/staff/working/tech-york/tech-commitment/</u>
Wellcome Sanger Institute	<u>https://www.techniciancommitment.org.uk/assets/action-plans-renewal/wsi-action-plan-2021-2024.pdf</u>
York St John University	<u>https://www.yorksja.ac.uk/media/content-assets/schools/staff/documents/YJSJ-Technical-Services-Commitment.pdf.pdf</u>

Table S2: Initial manually-curated categories and keywords list

Category	Keywords
Professional Registration & Accreditation	professional registration, accreditation, registered scientist, chartered scientist, rsci, csci, rscitech, advance he, hea fellowship, employer champion
Career Pathways & Progression	career pathway, career path, job families, progression, promotion, career development, job descriptions, regrading, career framework
Training & Skills Development	training, skills, cpd, staff development, upskilling, mentoring, coaching, secondment, job shadowing, workshops, apprenticeship
Visibility & Communication	visibility, communication, newsletter, webpage, website, social media, showcase, conference, networking, posters, blogs
Recognition & Awards	award, recognition, prize, nominate, celebrate, excellence awards
Research & Grant Support	research, grant, funding, ref, kef, tef, authorship, co-author, attribution, publication, pi, co-investigator
EDI & Community	edi, equality, diversity, inclusion, athena swan, race equality, underrepresented, outreach, community, wellbeing, mental health
Sustainability & Operations	sustainability, sustainable, leaf, efficiency, procurement, asset management, succession planning, workforce planning
Governance & Representation	committee, representation, steering group, governance, technician voice, decision making

Table S3: Final Categories and Keywords list with weighting in parentheses
(keyword_categories_4.json)

Category	Keywords (weights)
Apprenticeships, Internships, Placements	apprenticeship (5.0), apprenticeships (5.0), apprentice (5.0), trainee (1.0), traineeship (1.0), T-Level (5.0), placement (1.0), grow (1.0), apprenticeship levy (1.0), degree apprenticeship (1.0), secondment (1.0), job shadowing (1.0), shadowing (5.0)
Career Frameworks & Role Definition	framework (1.0), career framework (1.0), job families (1.0), HETT (1.0), job description (1.0), role profile (1.0), role descriptor (1.0), job classification (1.0), service redesign (1.0), job title (1.0), person specification (1.0), role mapping (1.0)
Career Pathways & Progression	career pathway (7.0), career progression (7.0), career development (7.0), toolkit (3.0), progression (3.0), PDR (5.0), appraisal (3.0), performance development review (5.0), regrading (5.0), promotion (5.0), career journey (7.0), pathway (3.0)
Data & Workforce Analysis	demographics (3.0), HESA (3.0), workforce (3.0), leavers (1.0), HR system (1.0), workforce data (1.0), PowerBI (1.0), dashboard (1.0), technician staff data (1.0), technician data (1.0), SOC Code (1.0)
EDI (Equality, Diversity & Inclusion)	EDI (1.0), equality (1.0), diversity (1.0), inclusive (1.0), Athena Swan (1.0), Race Equality Charter (1.0), gender balance (1.0), underrepresented (1.0), BAME (1.0), LGBTQ+ (1.0), neurodiverse (1.0), disabled (1.0), positive action (1.0)
Environmental Sustainability	environmental sustainability (1.0), green lab (1.0), LEAF (1.0), net zero (1.0), carbon footprint (1.0), waste reduction (1.0), decarbonisation (1.0)
Equipment and Asset Sharing	equipment sharing (1.0), asset sharing (1.0), shared resources (1.0), shared equipment (1.0), shared facilities (1.0), collaborative assets (1.0), equipment pool (1.0), asset management (1.0), equipment database (1.0), inventory (1.0)
External Collaboration & Partnerships	external company (1.0), external organisation (1.0), company (1.0), companies (1.0), ITSS (1.0), UKRI (1.0), external stakeholder (1.0), partnership (1.0), external network (1.0), external committee (1.0), industry (1.0), industry partner (1.0), sector (1.0), other institutions (1.0), partner (1.0), collaboration (1.0), institution (1.0), schools (1.0)
Funding for Technicians	budget (1.0), development fund (1.0), business model (1.0), funding (1.0), funding source (1.0), technician funding (1.0), training fund (1.0), travel fund (1.0), financial support (1.0), cost recovery (1.0)
Mentorship & Support	mentoring (5.0), mentorship (5.0), coaching (5.0), guidance (5.0), mentee (5.0), mentor (5.0), mentoring support (1.0), peer support (1.0), line manager support (1.0), buddy system (1.0), pastoral care (1.0)
Monitoring & Evaluation of TC	evaluate the impact (1.0), evaluate impact (1.0), evaluate progress (1.0), track progress (1.0), measure success (1.0), dashboard (1.0), metrics (1.0), RAG (1.0), self-assessment (1.0), evaluate (1.0), action plan (1.0), assess impact (1.0), progress report (1.0), review (1.0), monitor (1.0), track (1.0), assess (1.0), report (1.0)
Networking and Presenting	conference (5.0), networking (5.0), networking event (1.0), network (1.0), forum (1.0), knowledge exchange (5.0), community (1.0), presenting (1.0), poster (1.0), talk (1.0), presentation (1.0), seminar (3.0), symposium (3.0), showcase (3.0), away day (3.0), activity day (3.0)
Ongoing Visibility & Communication	website (1.0), newsletter (1.0), communication (1.0), visibility (5.0), webpage (1.0), page (1.0), outreach (1.0), twitter (3.0), X (3.0), LinkedIn (3.0), Instagram (3.0), Facebook (3.0), YouTube (3.0), TikTok (3.0), social media (3.0), blog (3.0), raise awareness (1.0), communication plan (1.0), staff profile (1.0), case study (1.0), podcast (3.0), profiles (3.0), showcase (3.0), stories (1.0), promote (1.0)

Operational Sustainability	asset management (1.0), succession planning (1.0), efficiency (1.0), procurement (1.0), workforce planning (1.0), resilience (1.0), business continuity (1.0), job security (1.0), fixed-term (1.0), contracts (1.0), funding models (1.0), succession (1.0)
Professional Registration & Accreditation	professional registration (1.0), Advance HE (1.0), accreditation (1.0), employer champion (1.0), Science Council (1.0), accredited (1.0), chartership (1.0), registered scientist (1.0), rsci (1.0), csci (1.0), rscitech (1.0), HEA fellowship (1.0), Engineering Council (1.0), professional body (1.0)
Publications	authorship (1.0), fair attribution (1.0), attribution (1.0), paper (1.0), acknowledgement (1.0), publication (1.0), journal (1.0), article (1.0), research output (1.0), author (1.0), co-author (1.0), CRediT taxonomy (1.0)
Recognition & Awards	nominate (1.0), nomination (1.0), award (1.0), prize (1.0), technician of the year (1.0), reward (1.0), recognition (1.0), celebrate (1.0), celebration (1.0), recognise (1.0), excellence awards (1.0), STAR award (1.0), Papin Prize (1.0), thank you (1.0), ceremony (1.0), ceremonies (1.0)
Recruitment & Onboarding	onboarding (1.0), recruitment (1.0), hiring (1.0), induction (1.0), exit interview (1.0), retention (1.0), interview process (1.0), interviews (1.0), interviewing (1.0), new starter (1.0), job advert (1.0), succession planning (1.0), entry routes (5.0), recruitment process (1.0), new staff (1.0)
Representation in Institutional Governance	representation (1.0), decision making (1.0), board (1.0), steering group (1.0), policy (1.0), committee (1.0), governance (1.0), management (1.0), strategic planning (1.0)
Research Assessments	Research Excellence Framework (1.0), TEF (1.0), Teaching Excellence Framework (1.0), REF (1.0), KEF (1.0), REF2029 (1.0), REF2028 (1.0), Knowledge Exchange Framework (1.0), research excellence (1.0), research culture (1.0), research impact (1.0), research quality (1.0), research assessment (1.0), research evaluation (1.0)
Research Funding applications	research funding (1.0), grant (1.0), funding application (1.0), costing (1.0), funding proposal (1.0), grant proposal (1.0), funding bid (1.0), co-investigator (1.0), PI (1.0), Principal Investigator (1.0)
Teaching	module (1.0), teaching (1.0), curriculum (1.0), learning (1.0), education (1.0), student support (1.0), assessment (1.0), pedagogy (1.0), student experience (1.0), demonstrating (1.0), learning delivery (1.0), module information (1.0)
Technician Leadership	leadership programme (7.0), leadership skills (7.0), leadership roles (7.0), leadership (3.0), strategic lead (5.0), team lead (5.0), manager (1.0), supervision (1.0), lead technician (5.0), technical lead (5.0), technical manager (5.0)
Technician Voice & Feedback	survey (1.0), feedback (1.0), evaluation (1.0), questionnaire (1.0), poll (1.0), consultation (1.0), stakeholder engagement (1.0), technician voice (1.0), focus group (1.0), town hall (1.0), gather feedback (1.0), staff survey (1.0), consult (1.0), technician input (1.0)
Training & Skills Development	CPD (1.0), HEaTED (1.0), Continuing Professional Development (1.0), training (1.0), skills audit (1.0), staff development (1.0), skills development (1.0), upskilling (1.0), reskilling (1.0), workshops (1.0), skills gap (1.0), training programme (1.0), development (1.0), learning (1.0), skill (1.0)
Wellbeing & Mental Health	mental health (1.0), wellbeing (1.0), stress (1.0), resilience (1.0), work-life balance (1.0), mental health first aid (1.0), counselling (1.0), mindfulness (1.0), wellbeing provision (1.0), mental health champion (1.0), staff welfare (1.0)

Table S4: Subcategories with keywords and count scores.

Subcategory	In Main Category	Count	Keywords
Community Building	Networking and Presenting	50	community, communities, practice, groups, connect
Management & Senior Roles	Technician Leadership	53	management, leaders, senior, managers, lead
Internships & Placements	Apprenticeships, Internships, Placements	54	internship, internships, placement, placements, work, experience, student
EDI Initiatives & Strategy	EDI (Equality, Diversity & Inclusion)	54	edi, equality, diversity, inclusion, strategy, action, plan
Showcasing & Presenting	Networking and Presenting	56	present, presenting, showcase, opportunities, posters, talks
Engagement with External Partners	External Collaboration & Partnerships	60	external, partners, collaborate, collaboration, stakeholders, industry
Securing Internal Funding	Funding for Technicians	62	funding, fund, budget, resource, financial, support, central
Coaching & Guidance	Mentorship & Support	64	coaching, coach, guidance, buddy, peer
Data Collection & Mapping	Data & Workforce Analysis	65	data, mapping, workforce, collect, analysis, understand
Committee Representation	Representation in Institutional Governance	66	representation, committee, committees, groups, meetings
Mentoring Schemes	Mentorship & Support	70	mentoring, mentor, mentors, scheme, schemes, programme,
Promotion of Registration	Professional Registration & Accreditation	71	professional, registration, promote, encourage, promotion, ,
Progress Monitoring & Reporting	Monitoring & Evaluation of TC	74	monitor, progress, evaluation, report, track, measure,
Role Profiles & Job Families	Career Frameworks & Role Definition	86	role, roles, profiles, job, families, define, definition
Apprentice & Trainee Schemes	Apprenticeships, Internships, Placements	91	apprenticeship, apprenticeships, trainee, scheme, schemes
Internal Communications	Ongoing Visibility & Communication	98	communications, comms, newsletter, internal, channels, bulletin
Website & Online Profiles	Ongoing Visibility & Communication	100	website, online, profiles, web, webpages, digital
Career Development & Opportunities	Career Pathways & Progression	115	career, development, opportunities, develop
Awards & Recognition Schemes	Recognition & Awards	123	awards, award, recognition, scheme, schemes, ceremony
Training Needs & Skills Audit	Training & Skills Development	145	training, skills, development, needs, audit, analysis
Promotion & Progression Routes	Career Pathways & Progression	151	promotion, progression, pathway, pathways, advancement

Promotion & Case Studies	Ongoing Visibility & Communication	152	promote, promotion, visibility, case, studies, highlight, showcase
Networking Events & Conferences	Networking and Presenting	165	network, networking, events, conference, forum, symposium
Action Plan Management	Monitoring & Evaluation of TC	187	action, plan, review, delivery, actions, objectives